



FINAL PROJECT REPORT - MALEKULA

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AAM Project Number: 20076A

Project Name: Acquisition of LiDAR and Imagery Derived Products for Vanuatu

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1.0 Introduction

AAM was engaged by Geoscience Australia to collect topographic and bathymetric data required to assess key vulnerabilities and risks, formulate adaptation strategies and plan, in addition to mainstreaming adaptation into decision making along the Vanuatu coast.

Acquisition was undertaken between September 11th and November 26th 2012. Rain, low cloud and other weather related challenges were faced in this aerial LiDAR survey.

AAM deployed its Optech ALTM Pegasus for this project. This sensor is capable of detecting multiple returns, with a minimum of 4 potential returns for each outbound laser pulse as well as recording the intensity of each return.

The planned point density for the survey was an average density of 3.0pts/m² for the topographic LiDAR capture.

The vertical accuracy for this data set is +/-0.30m, and the horizontal accuracy is +/-0.80m. This data is presented in terms of ITRF2008 UTM zones 58/59. The vertical data is presented in terms of ITRF2008 Ellipsoidal and MSL (2012).

DATA IN THIS FINAL REPORT RELATES TO THE MALEKULA DATASET WHICH WAS TOPOGRAPHIC ONLY. FURTHER FINAL REPORTS WILL BE PRODUCED FOR EFATE AND ESPIRITO SANTO.

This *Malekula Project Report* addresses the following topics as specified in the contract:

1. how each of the contract specifications has been met,
2. a statement of consistency with any specified standards,
3. results of independent accuracy tests,
4. metadata statements
5. extra-ordinary issues that may have affected the nature or delivery of the project
6. other project information of note.

2.0 Contract Specifications Confirmed

2.1 Category of Elevation Capture and Laser Flight Settings

The survey undertaken was ICSM Category 1:

- Absolute vertical accuracy: 1 sigma 0.30m or better in areas of open, flat terrain.
- Absolute horizontal accuracy: 1 sigma 0.80m or better in areas of open, flat terrain.

The category of elevation is assigned based on LiDAR system specifications and settings used for project acquisition. The following Table shows the LiDAR system settings used for acquisition.

Laser Flight Settings		
Description	Malekula	Units
Sensor ID	Optech ALTM Pegasus	
Flying Height	700	m
Swath Overlap	30	%
Swath Width	509	m
Half Scan Angle	20	Degrees
Scan Frequency	40	Hz
Maximum Number of Returns	4	
Footprint Size	0.14	m
Pulse Rate Frequency	250	KHz
Estimated Horizontal Accuracy	0.8	m
Estimated Vertical Accuracy	0.3	m
Average Point Spacing	0.32	Pts/m ²
Average Density of Ground Strikes	7.34	Pts/m ²

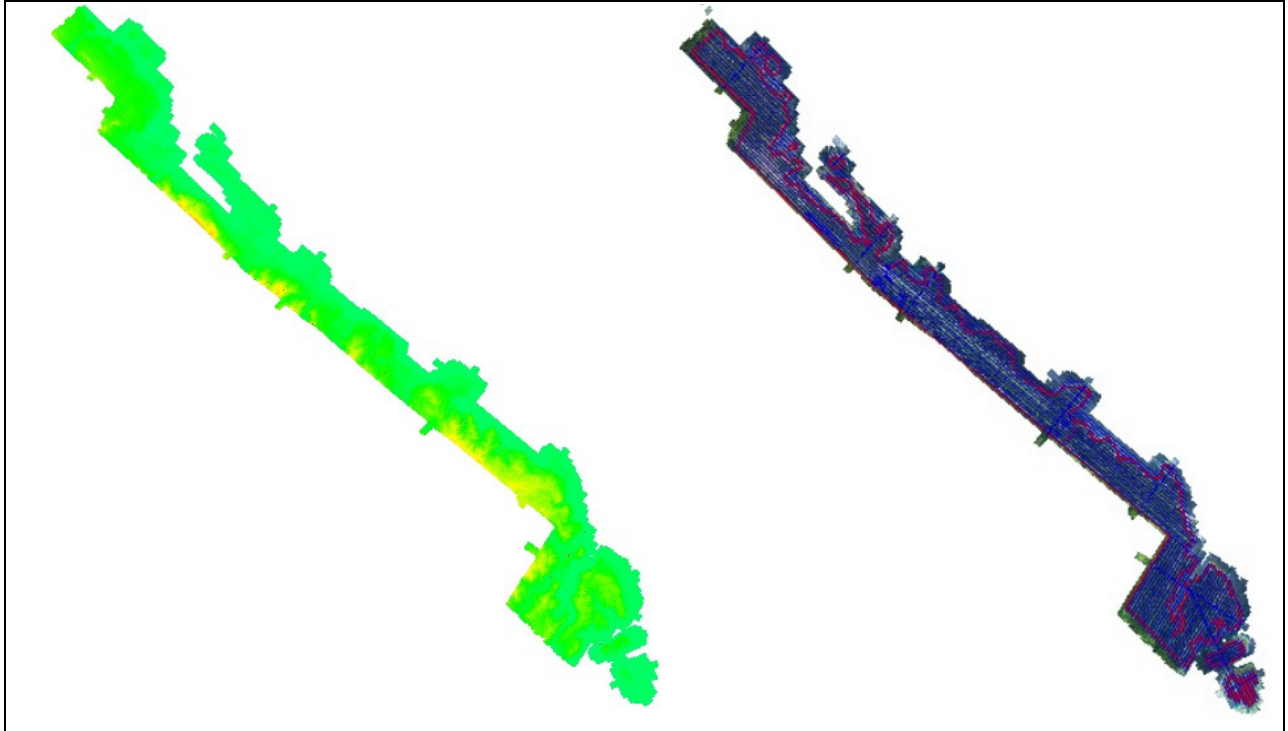
The Figure below displays the LiDAR system accuracy specifications as presented by Optech for the ALTM Pegasus.

Parameter	Specification
Operational envelope ^{1,2,3,4}	150-5000 m AGL, nominal
Laser wavelength	1064 nm
Horizontal accuracy ²	1/5,500 x altitude, 1 σ
Elevation accuracy ²	<5-20 cm, 1 σ
Effective laser repetition rate	Programmable, 100-500 kHz
Position and orientation system	POS AV™ AP50 (OEM)
Scan width (FOV)	Programmable, 0-75°
Scan frequency ⁵	Programmable, 0-140 Hz (effective)
Sensor scan product	800 maximum
Beam divergence	0.25 mrad (1/e)
Roll compensation	Programmable, $\pm 37^\circ$ (FOV dependent)
Vertical target separation distance	<0.7 m
Range capture	Up to 4 range measurements, including 1st, 2nd, 3rd, and last returns
Intensity capture	Up to 4 intensity returns for each pulse, including last (12 bit)
Image capture	5 MP interline camera (standard); 60 MP full frame (optional)
Full waveform capture	12-bit Optech IWD-2 Intelligent Waveform Digitizer (optional)
Data storage	Removable solid state disk SSD (SATA II)
Power requirements	28 V, 800 W, 30 A
Dimensions and weight	Sensor: 630 x 540 x 450 mm; 65 kg; Control rack: 650 x 590 x 490 mm; 46 kg
Operating temperature	-10°C to +35°C
Relative humidity	0-95% non-condensing

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2.2 Coverage

The area of interest on Malekula covers an area of approximately 294sqkm.



Malekula Overview – LiDAR Elevation Data and Imagery Data

2.3 Horizontal Datum

The survey was coordinated in terms of ITRF2008.

2.4 Vertical Datum

For data presented in ITRF2008 all elevation data has been presented in Mean Sea Level (MSL) 2012. Ellipsoidal data has also been presented.

In order to verify the accuracy of the EGM2008 model or any adjustment to the vertical heights, a number of tidal stations were recovered across the Vanuatu Island chain in order to establish the height difference between the 1'x1' EGM2008 model and local Mean Sea Level.

Examination of existing tidal records revealed that the tidal regime is primarily semi-diurnal with a diurnal inequality evident. This results in a tidal regime that has two highs and two low waters a day but one of these cycles will be a higher high and low water.

During processing of all the GPS benchmarks, orthometric heights were calculated using the 1'x1' EGM2008 mark. As a result it was possible to calculate the ellipsoid to geoid (MSL) value for these points. These points were then compared with the derived values.

The below table demonstrates the relationship between EGM2008 and MSL for all of the tide gauge sites across the three islands of Efate, Espiritu Santo and Malekula.

MSL heights on wharf survey marks						
Wharf Control Point (tide gauge connected)		Phase 1 Resolved Heights (GPS only)			Phase 2 Adopted Heights (Tide gauge)	
		Ellipsoid Height (GRS80)	Geoid Separation (EGM2008)	Orthometric height (EGM2008)	MSL (measured)	Geoid Separation (corrected)
Island	ID	(A)	(B)	(C)	(D)	(E)
Efate	povi	66.872	63.585	3.287	2.550	66.872
	emua	68.067	65.624	2.443	1.642	68.067
Malekula	lili	66.077	63.756	2.321	1.294	66.077
Santo	saw1	67.635	64.855	2.780	1.750	67.635
	saw2	67.573	64.854	2.719	1.688	67.573

In order to validate the MSL values calculated through the use of short period observations a comparison was made of the reading of Port Vila against longer period observations held by the South Pacific Sea Level Climate Monitoring Project (SCSLCMP). As all other island MSL values were in close agreement with Port Vila MSL, it was felt that this comparison validates all of the results as shown below.

Comparison with Annual MSL Values	Mean Sea Level (m)	Observed Mean Sea Level (m)	Difference
2010	0.899	0.904	+0.005
2011	0.919		-0.015
2012 (to Oct)	0.905		+0.001
2010-2012	0.908		-0.004

The results obtained are a good indicator of the differences between the EGM2008 and local orthometric height datum and should be used to correct the data. It is assessed that the MSL values are correct to +/- 0.004m with only average differences of 0.05m seen during the period of observations and analysis.

2.5 Map Projection

The survey data is presented in terms of ITRF2008 UTM zone 58.

2.6 Survey Control

The primary survey control point (GPS Base stations) was established at Luganville Airport on Espiritu Santo. Additional control marks were established on Malekula at Lemap and the Litz Litz wharf.

The central base stations were coordinated using AUSPOS (24 hour periods, multiple days) and checked against a number of surrounding State Survey Control Points.

All reference and check point surveys were coordinated relative to the nearest base station. All base station, reference points, and check points were coordinated against local survey marks.

Summary of the datum used throughout the project is below:

Horizontal:

The horizontal datum was established directly from AUSPOS for each base station. The Base stations accuracies were subsequently checked by control survey from established coordinated survey marks.

Vertical:

Ellipsoid - Established directly from AUSPOS as per the horizontal datum.

MSL - Each base station, reference point survey, and check point survey was linked to surrounding MSL marks, thereby providing them a "local " MSL datum in their immediate vicinity.

2.7 Data Tiling and File naming

Data is provided in tiles 1km by 1km to the following filenaming convention:

Intensity and RGB Imagery

Naming Convention for LiDAR intensity or other forms of imagery: <i>ProjectNameYYYY-INT-GSD xxxyyyyy zz wwwwww hhhh.ecw</i>		
ProjectName	<i>EfateLidar</i>	A meaningful description of the total survey area of interest. Do not use “_” as part of the Project Name
YYYY	<i>2012</i>	Year of survey
INT/RGB	<i>-INT</i>	Intensity image file identifier. Use RGB for 3 band natural colour imagery or RGBI for 4 band infrared
GSD	<i>-002</i>	Ground sampling distance or resolution of image in metres.
xxxxyyy	<i>_4806558 (480,000mE) (6558,000mN)</i>	Easting and northing value (whole kilometre) of the south- west corner of the tile. A single “_” must be used to separate the remaining file name components.
zz	<i>_56</i>	UTM zone of the file
wwwwww	<i>_0020</i>	Width of the dataset or tile in whole kilometres
hhhh	<i>_0050</i>	Height of dataset or tile in whole kilometres
<i>For example: EfateLidar2009-INT-002_4806558_56_0020_0050.ecw</i>		

Unclassified LiDAR

Naming Convention for LiDAR point clouds: <i>ProjectNameYYYY-UNC-DAT-SWT xxxyyyyy zz wwwwww hhhh.las</i>		
ProjectName	<i>EfateLidar</i>	A meaningful description of the total survey area of interest. Do not use “_” as part of the Project Name
YYYY	<i>2012</i>	Year of survey
UNC	<i>-UNC</i>	Unclassified point cloud. Fully calibrated and adjusted to specified datum
DAT	<i>-ELL</i>	Ellipsoidal heights (PNG94)
SWT	<i>-1..n</i>	Swath number (1 file per swath)
xxxxyyy	<i>_4806558 (480,000mE) (6558,000mN)</i>	Easting and northing value (whole kilometre) of the south- west corner of the tile. A single “_” must be used to separate the remaining file name components.
zz	<i>_56</i>	UTM zone of the file
wwwwww	<i>_0002</i>	Width of the tile in whole kilometres
hhhh	<i>_0002</i>	Height of the tile in whole kilometres
<i>For example: EfateLidar2009-RAW-ELL-001_4806558_56_0002_0002.las</i>		

Classified LiDAR

Naming Convention for LiDAR point clouds:		
<i>ProjectNameYYYY-CL-DAT xxxxyyy zz wwww hhhh.las</i>		
ProjectName	<i>EfateLidar</i>	A meaningful description of the total survey area of interest. Do not use “_” as part of the Project Name
YYYY	<i>2012</i>	Year of survey
CL	<i>-C2</i>	classification level.
DAT	<i>-ELL or MSL</i>	Specified vertical datums. Ellipsoidal (ELL) or Orthometric (MSL)
xxxxyyy	<i>_4806558 (480,000mE) (6558,000mN)</i>	Easting and northing value (whole kilometre) of the south- west corner of the tile. A single “_” must be used to separate the remaining file name components.
zz	<i>_56</i>	UTM zone of the file
wwww	<i>_0002</i>	Width of the tile in whole kilometres
hhhh	<i>_0002</i>	Height of the tile in whole kilometres
<i>For example: EfateLidar2009-C3-MSL_4806558_56_0002_0002.las</i>		

ESRI Grids

Naming Convention for tiled UTM ESRI GRIDS:		
<i>txxyyyssppp</i> (in addition to folder structure above with UTM GRIDS stored in separate folders, with all projection information defined.)		
t = surface type.	<i>e</i>	Surface type • s – digital Surface model (DSM) • e – digital Elevation model (DEM) • f - canopy Foliage model (CFM) • c - Canopy elevation model (CHM) • h – Hydro digital elevation model (DEMH) • b – Bathymetry • m – Bathymetry and terrain elevations • t – Derived terrain variables (add as necessary)
xxxxyyy	<i>6458595 (645,000mE) (8,595,000mN)</i>	• Easting and northing value (whole kilometre) of the south- west corner of the tile.
ss	<i>01</i>	Tile size (km) (square tile) • 01 – one kilometre • 02 – two kilometre • 05 - five kilometre • 10 – 10 kilometre • _5 (represents half a kilometre)
ppp	<i>001</i>	Ground sampling distance (GSD) or pixel size • 0_5 - half a metre • 001 – one metres • 002 – two metres etc
<i>For Example: e645859501001</i>		

Naming Convention for Mosaic (UTM) ESRI GRIDS: <i>xxxxxxxxyppp</i> <i>(in addition to folder structure above with UTM GRIDS stored in separate folders, will all projection information defined.)</i>		
t = surface type.	<i>e</i>	Surface type • s – digital Surface model (DSM) • e – digital Elevation model (DEM) • f - canopy Foliage model (CFM) • c - Canopy elevation model (CHM) • h – Hydro digital elevation model (DEMH) • b – Bathymetry • m – Bathymetry and terrain elevations • t – Derived terrain variables (add as necessary)
xxxxxxx	<i>efate</i>	A meaningful description of the total survey area and or sensor, dataset version etc.
yy	<i>12</i>	Year of Survey
ppp	<i>010</i>	Ground sampling distance (GSD) or pixel size in metres (UTM) UTM • 0_5 - half a metre • 001 – one metres • 002 – two metres etc
<i>For Example: eEefate07010</i>		

Naming Convention for all other UTM files:		
<i>ProjectNameYYYY-SSSS-PPPP-GSD xxxyyyyy zz www hhhh.asc</i>		
ProjectName	<i>EfateLidar</i>	A meaningful description of the total survey area of interest. Do not use "_" as part of the Project Name
YYYY	<i>2012</i>	Year of survey
SSSS-PPPP	<i>-DEM-GRID</i>	Surface type. • DSM • DEM • HDEM • CHM • CFM • Bathymetry (BAT) • Mixed (MIX). Bathymetry and terrain elevations • TTT (Other terrain variables e.g. slope (SLP). Add as necessary. Product type • Mass points (MASS) • Breaklines (BRK) • TIN (TIN) • GRID (GRID) • Contours (CON) • Cross Sections (CROSS) • Imagery (BIL, TIF, IMG, ECW etc) • Other Use additional field width and more characters if required.
GSD	<i>-010</i>	Ground sampling distance or resolution of product where appropriate. Where GSD is not required producers can extend the surface type and product description field.
xxxxyyy	<i>_4806558 (480,000mE) (6558,000mN)</i>	Easting and northing value (whole kilometre) of the south- west corner of the tile. A single "_" must be used to separate the remaining file name components.
zz	<i>56</i>	UTM zone of the file
www	<i>_0020</i>	Width of the dataset or tile in whole kilometres
hhh	<i>0050</i>	Height of dataset or tile in whole kilometres
ext		File extension according to format conventions • LAS • xyz ascii format for easting, northing, elevation, intensity • asc – ESRI ascii GRID format • shp • dxf etc
<i>For example: EfateLidar2009-DEM-GRID-010_4806558_56_0020_0050.asc</i>		

Primary LiDAR Point Cloud Data

The LAS files have been provided to ASPRS V1.2 Standards.

LAS file point classifications levels are formatted to comply with ASPRS Standard LiDAR Point Classes.

1	Unclassified
2	Ground
3	Low Vegetation (<1m)
4	Medium Vegetation (>1m – 2m)
5	High Vegetation (>2m)
6	Building
7	Low Point (noise)
8	Model Key-point (mass point)
9	Water
10	Reserved for ASPRS Definition

All returns and intensity has been retained. Overlap data has been retained. LAS files retain all relevant capture data including: point source ID and all flight path, sensor, GPS and processing system data. Refer to Section 2.7 data tiling and file naming for data tiling and file naming specifications.

2.8 Digital Elevation Model (DEM)

ArcGIS Terrains were created from ground classified LIDAR data. One metre ESRI ArcGIS binary grids were derived using a customised ArcGIS macro.

2.9 Digital Surface Model (DSM)

ArcGIS surfaces were created from non-ground classified LIDAR data. One metre ESRI ArcGIS binary grids were derived using a customised ArcGIS macro.

2.10 Canopy Elevation Model (CEM)

A canopy elevation model was produced as two metre ESRI ArcGIS binary grids using ArcGIS with the following methodology:

- All ground surface derived using natural neighbours interpolation [G] (*Floating Point*)
- First return vegetation surface derived using natural neighbours interpolation [V1] (*Floating Point*)
- Height in metres above ground of first vegetation returns $[V1] - [G] = [H]$ (*Floating Point*)
- Mask of all vegetation ALS returns with 1m buffer, reclassified to 0 outside mask and NoData inside [M] (*Integer*)
- Mosaic of canopy height surface and vegetation mask $[H] + [M] = \text{CEM}$ (*Floating Point*)

2.11 Foliage Cover Model (FCM)

A foliage cover model was produced as 10 metre ESRI ArcGIS binary grids using ArcGIS with the following methodology:

- Count of all vegetation returns per 10m cell [V] (*Integer*)
- Count of total ALS returns per 10m cell [A] (*Integer*)
- Proportion of all vegetation to total ALS returns expressed as a percentage $([V] / [A]) * 100 = \text{FCM}$ (*Floating Point*)

2.12 Contours

Contours were generated in ESRI Shape file format with 0.5m and 5m intervals.

2.13 Imagery

Imagery has been generated in ECW and TIFF format to the capture extents. Please see details of camera specifications below:

DiMAC cameras bring high-quality imagery to a medium format ■ Patented true Forward Motion Compensation technology provides crisp, blur-free imagery (fly lower, faster and longer) ■ Temperature stabilization ■ Rigorous geometric and radiometric calibration ■ Wide choice of lenses ■ 1-year comprehensive manufacturer's warranty with 24/7 support		ALTM Multi-Sensor Integration Package ■ Custom-engineered multi-sensor mount ■ Fully integrated lidar and photo mission planning capability ■ Fully integrated sensor control and monitoring capability ■ All required interconnect cables ■ Full calibration report with lidar/camera boresight values ■ Planning, operations, and data-processing training	
Parameter		Specification	
Camera Module (CM)			
Area sensor		60 MP full-frame CCD 8,984 x 6,732 pixels (effective) 6 µm x 6 µm pixels 53.9 x 40.4 mm (effective)	
True Forward Motion Compensation (FMC)		Solid-state electro-mechanical	
Lenses		50 mm / 70 mm / 120 mm / 210 mm	
Shutter		Electro-mechanical iris mechanism	
Filter		1/125 to 1/500 sec., f-stops: 4, 5.6, 8, 11, 16	
Image output		Standard size RGB and IRC interchangeable filters	
		8 or 16 bits per channel 24-bit RGB: 180 MB 48-bit RGB: 360 MB	
Capture rate		2.5 sec.	
Resolution (GSD)		2 cm to 1 m / <1 inch to 3.3 feet	
IT Cube (ITC)			
Control & acquisition computers		PC/104 RoHS-compliant small form-factor embedded computers with minimum: Intel® Core™ Duo LV2400 CPU, 1-GB RAM 4-GB flash disk local storage IEEE 1394 Firewire interface	
Removable storage units		500-GB pressurized hard drives - 8,000 images typically	
Dimensions (H x W x D)		250 x 250 x 250 mm / 10 x 10 x 10 in	
Weight		15 kg / 33 lbs	
Input voltage (DC)		24-28 V	
Current draw		5 A	
Software			
DiOPS		Real-time camera operations and configuration management	
CaptureOne		Image development	

2.14 Project Plan

Prior to data capture commencement, a Project Plan was submitted to and approved by Geoscience Australia. This Project Plan details work breakdown structure, agreed data capture plans, project milestones and delivery schedules, progress reporting schedules.

2.15 Pre-survey Quality Assurance Plan

Prior to the commencement of aerial LiDAR data capture, AAM prepared and submitted to Geoscience Australia a Quality Assurance Plan that conformed to an identified management system and generally complies with ISO 9001. The Plan addressed the organisation and management of the project, work procedures, environmental considerations, safety and risk control and test procedures.

The Plan also detailed the procedures to be used in verifying that the deliverables meet the required specification including:

- the procedures and methodologies to be used to verify that the deliverables meet the required specifications.
- Details of proposed calibration checks and methodology to be used to establish both reference stations and ground test sites.

2.16 Post-survey Spatial Accuracy Report

Following successful capture of the aerial data a Post Survey Spatial Accuracy report was submitted and approved by Geoscience Australia.

Observed flight runs were provided in ESRI shape file format.

All reference and check points used to verify accuracy were provided in ESRI shape file format.

For analysis of ground comparisons refer to Section 3.0. Accuracy Assessment.

2.17 GPS Base Station Data

GPS data for all occupations of base-stations has been supplied in RINEX format (Receiver Independent Exchange Format)

3.0 Accuracy Assessment

Accuracy assessment was undertaken in two phases. Initially a post-survey reference site assessment was conducted to analyse the overall accuracy of the transformed lidar ground strikes relative to MSL. Additional check points were then collected to independently assess the accuracy of the classified data according to different land cover types. In total 424 field survey points were collected across the Malekula lidar capture area.

The independent ground survey was not undertaken in an appropriate method to allow for classification accuracy assessment of landcover classes in the form of an error matrix. The survey did however, allow for a detailed assessment of vertical accuracy across different land cover classes. ID photos were only collected for each survey site and not individual points.

3.1 Project Design Accuracy

Project specifications and technical processes were designed to achieve data accuracies as follows:

Project Design Accuracy			
	Measured Point	Derived Point	Basis of Estimation
Vertical data		0.15m	Project Design
Horizontal data	< 0.13m		System specifications
Test points	0.05m		Survey methodology used

Notes On Expected Accuracy

- Values shown represent standard error (68% confidence level or 1 sigma), in metres.
- "Derived points" are those interpolated from a terrain model.
- "Measured points" are those observed directly.
- Accuracy estimates for terrain modelling refer to the terrain definition on clear ground. Ground definition in vegetated terrain may contain localized areas with systematic errors or outliers which fall outside this accuracy estimate.
- Laser strikes have been classified into "ground" and "non-ground", based upon algorithms tailored for major terrain/vegetation combinations existing in the project area. The definition of the ground may be less accurate in isolated pockets of dissimilar terrain/vegetation combinations.

3.2 Reference Site Analysis

Test points were distributed in 4 groups across the Malekula Island and located on open clear ground. Comparison of the field test points with elevations interpolated from measured data are summarised in the table below.

Reference Point Site Vertical Accuracy – LiDAR Data			
Reference Point Site	Mean Difference (m)	Std Deviation (m)	RMS (m)
TIP09_MSL_031212	0.066	0.050	0.082
TIP09_MSL_031212	0.040	0.040	0.050
TIP09_MSL_031212	0.068	0.032	0.075
TIP09_MSL_031212	0.033	0.024	0.039

3.3 Check Point Analysis

For the purposes of independent accuracy assessment check point measurements were taken throughout the project area. Check points were collected for each land cover type across the project extent. The Table below shows the number of check points for each class.

Area	Landcover Class	No. of Points
Malekula	Grass	52
	Grass (0.01m high)	72
	Grass (~0.01m high)	30
	Grass (~0.5m high)	12
	Grass and shrubs (~0.5m high)	7
	Shrubs bushes (~3m high) slight slope	12
	Shrubs bushes (~5m high)	10
	Under trees	26
	Under trees (~5m high)	7
	Under trees (~5m high) slight slope	18
	Under trees (~10m high) thick canopy	8
	Under high trees (up to 20m high)	27
	Under trees (~20m high)	33
	Under shrubs and low trees.	15

Comparison of the field check points with elevations interpolated from the LiDAR and DEM data resulted in the accuracies shown in the following tables for the landcover classes:

Vertical Accuracy – LiDAR Data					
Area	Landcover Class	No. of Points	Mean Difference (m)	Std Deviation (m)	RMS (m)
Malekula	Grass	52	+0.074	0.035	0.082
	Grass (0.01m high)	72	-0.021	0.071	0.074
	Grass (~0.01m high)	30	-0.015	0.023	0.027
	Grass (~0.5m high)	12	+0.120	0.177	0.208
	Grass and shrubs (~0.5m high)	7	+0.243	0.158	0.284
	Shrubs bushes (~3m high) slight slope	12	+0.035	0.079	0.083
	Shrubs bushes (~5m high)	10	+0.015	0.052	0.052
	Under trees	26	+0.005	0.105	0.103
	Under trees (~5m high)	7	+0.081	0.090	0.116
	Under trees (~5m high) slight slope	18	-0.029	0.048	0.055
	Under trees (~10m high) thick canopy	8	-0.032	0.054	0.060
	Under high trees (up to 20m high)	27	-0.107	0.052	0.118
	Under trees (~20m high)	33	+0.011	0.079	0.079
	Under shrubs and low trees.	15	-0.007	0.097	0.094

Vertical Accuracy Comparison with Data Specification – LiDAR Data						
Area	Landcover Class	Within ±0.30m	Outside ±0.30m	Total	% Within ±0.30m	% Outside ±0.30m
Malekula	Grass	52	0	52	100	0
	Grass (0.01m high)	72	0	72	100	0
	Grass (~0.01m high)	30	0	30	100	0
	Grass (~0.5m high)	10	2	12	83	17
	Grass and shrubs (~0.5m high)	4	3	7	57	43
	Shrubs bushes (~3m high) slight slope	12	0	12	100	0
	Shrubs bushes (~5m high)	10	0	10	100	0
	Under trees	26	0	26	100	0
	Under trees (~5m high)	7	0	7	100	0
	Under trees (~5m high) slight slope	18	0	18	100	0
	Under trees (~10m high) thick canopy	8	0	8	100	0

	Under high trees (up to 20m high)	27	0	27	100	0
	Under trees (~20m high)	32	1	33	97	3
	Under shrubs and low trees.	15	0	15	100	0

4.0 Metadata

A complete metadata statement consistent with the current ANZLIC standard has been provided for each data product supplied. Metadata documents have been provided in Adobe Reader (.pdf) format. Further for every file in each data product, an .xml document with a complete metadata statement consistent with the current ANZLIC standard has been provided.

4.1 General Capture Specifications

MetaData Elements	
Characteristic	Description
Device Name	Optech ALTM Pegasus
IMU Used	SAGEM FMU
Acquisition Start Date	29/09/2012
Acquisition End Date	09/10/2012
Number of Runs	147
Number of Cross Runs	14
Flight Direction	Variable
Flying Height (AGL)	700m
Half Scan Angle	20
Swath Width	509
Sidelap	30%
Average Point Spacing	0.32
Laser Pulse Rate	250KHZ
Laser Pulse Mode	Single
Laser return Types	1 st , 2 nd , 3 rd and Last
Laser Intensity	Supplied on all returns
Laser footprint size	0.14m
DEM output pixel size	1m
DSM output pixel size	1m
CHM output pixel size	2m
FCM output pixel size	10m
Horizontal Datum	ITRF2008
Vertical Datum	ITRF2008 MSL (2012),
Map Projection	UTM zones 58
System Calibration Certification	20 th June 2012

4.2 Source Data

Source Data				
Item	Source	Description	Ref No	Date
Laser System	AAM	ALTM-Pegasus	20076A	11/09/12-26/11/12
GPS Base Data	AAM	Static GPS	20076A	11/09/12-26/11/12
Base Stn Coords	AAM	Total Station	20076A	11/09/12-26/11/12
Field Survey Data	AAM	RTK & Total Station	20076A	11/09/12-26/11/12

4.3 Reference Systems

	Horizontal	Vertical
Datum	ITRF2008	MSL
Projection	UTM Zone 59	N/A
Geoid Model	N/A	Tide Gauge Calculated
Base Station 1	Espiritu Santo Airport 738029.598E 8284336.503N	Espiritu Santo Airport 42.791 RL
Base Station 2	Lemap 797119.842E 8179126.596N	Lemap 5.986 RL
Base Station 3	Wharf 3 (Litz Litz) 761373.858E 8217066.306	Lemap 1.294 RL

4.4 Limitations of Data

The definition of the ground under trees and over wetland areas may be less accurate.

5.0 Extraordinary Issues

There have been no extraordinary issues associated with this project.

6.0 Other Project Information of Note

6.1 ANZLIC Metadata Statement

To be generated from NEDF/ANZLIC metadata tool

6.2 Post Survey Spatial Accuracy Report (as supplied December 2012)

Introduction

The acceptance of this report by Geoscience Australia is required before point classification and other product derivation is to proceed. The elements of this deliverable are to include

- Observed flight runs are to be provided in ESRI shape file format. The shape file should contain the following attributes: RUN, DATE and TIME for each run.
- All survey control coordinates, site ID and checkpoint comparisons in both EXCEL spreadsheet and ESRI shape file formats.
- Details of system calibration checks.
- Results of relative (flight run) matching and details of any adjustment made.
- Source of primary ellipsoidal control.
- Results of vertical and horizontal accuracy validation.
- Details of ellipsoid to orthometric corrections applied including any final adjustment to MSL supplemental to the standard Geoid correction.
- Relationship of the collected bathymetric LiDAR data to previously collected topographic LiDAR data at TIP sites including ESRI grid representing the differences.
- Digital photographs of all survey and check sites, with the site ID included in the file name. The bearing of the photo direction should also be indicated.

1.0 Observed Flight Runs

Observed flight runs are to be provided in ESRI shape file format. The shape file contains the following attributes: RUN, DATE and TIME for each run.

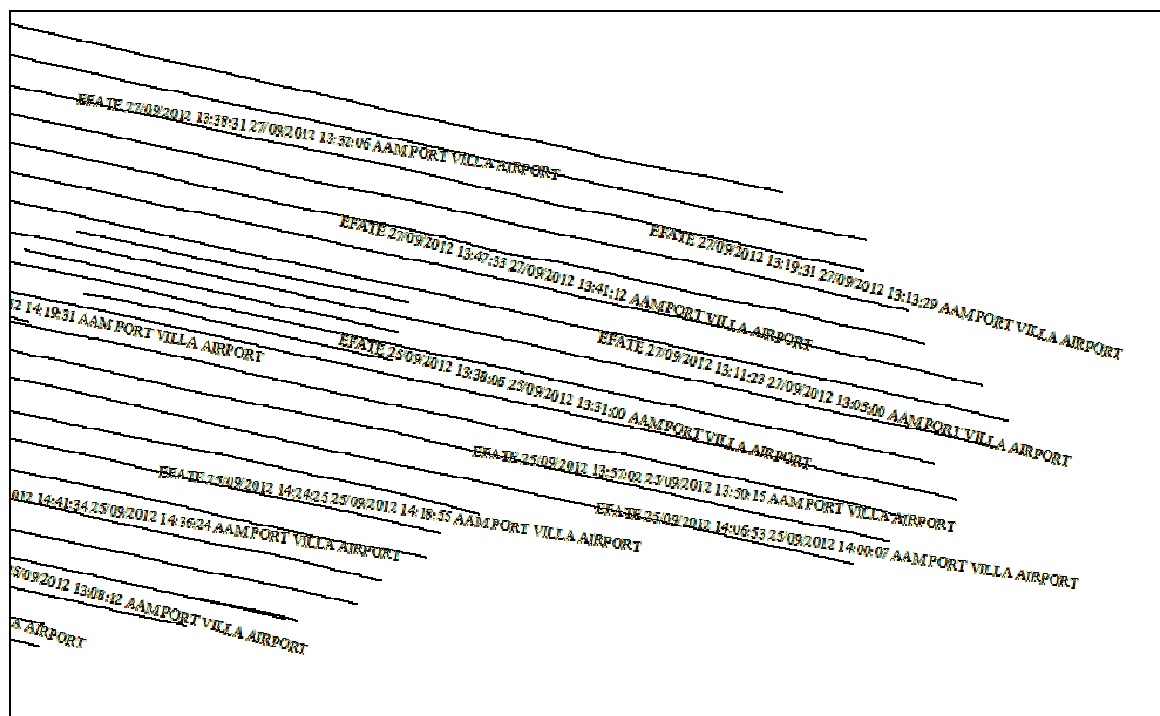


Fig 1.1 – Diagram showing example of observed flight runs for Efate provided in ESRI SHP file.

2.0 Survey Control Check Points

All survey control and/or check points used to verify accuracy are provided in ESRI shape file format.

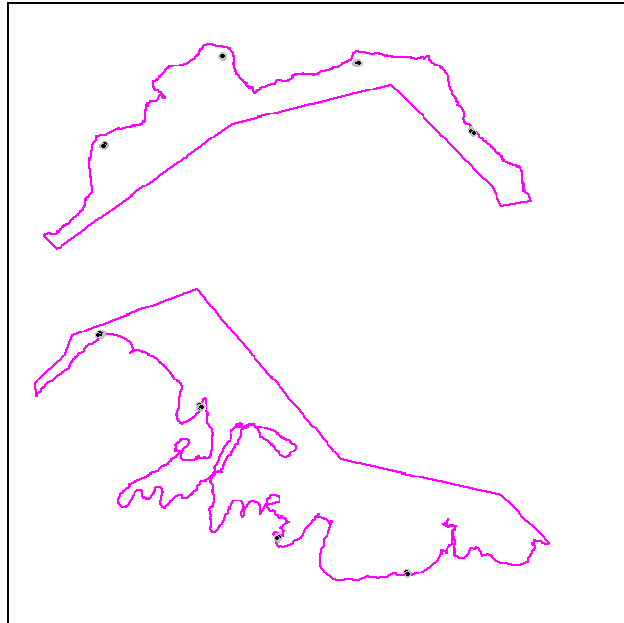


Fig 2.1 – Test Point Locations on Efate

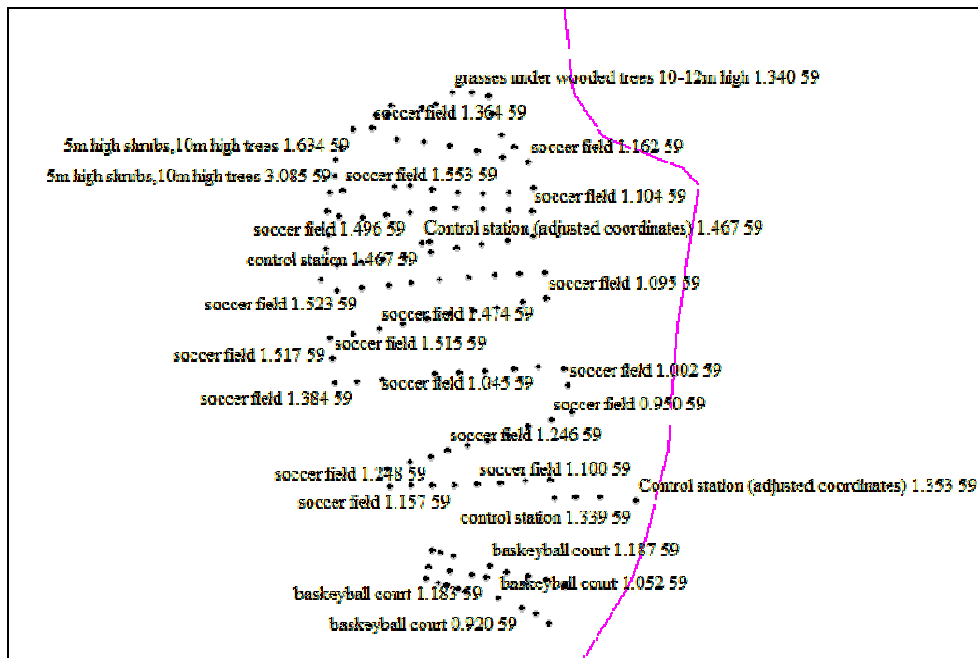


Fig 2.2 – Example of Test Point Locations for one area on Efate.

3.0 Results of Horizontal & Vertical Calibration Checks

Results of horizontal and vertical calibration checks, both relative (flight runs) and absolute (ground truth), including details of any adjustments made.

3.1 Base Station Analysis:

The Vanuatu Land Survey Department provided a number of control stations along the coastline adjacent to the mapping polygon.

At the Port Vila (Efate) and Luganville (Santo) International Airports, AAM established Primary and Secondary survey control points to observe base GPS data when the Lidar was being captured. These survey control points are permanent structures for future reference and were tied into ground survey network by static GPS.

After locating the permanent survey control provided by the Land Survey Department, new survey control marks were required at each test point location (TIP).

The coordinates for the permanent survey control points provided by the Land Survey Department were of unknown history. It was decided all primary control marks (permanent) would be measured by long static observations and processed using Geosciences Australia Auspos solution.

In most situations 6 hours of data was observed to generate the primary control coordinates. This process meant that on each island there were a number of primary controls established and used in conjunction with the previously established airport control stations on Efate and Santo.

Each network consists of static baselines observed on each island, connecting the adopted Primary Control Marks and the secondary control marks at each TIP site. Each network was adjusted using Leica Geo Office software.

The network included all observed baselines and primary station coordinates to do the adjustment. Each primary control station was assigned fixed point status and constrained horizontally and vertically to according to the Auspos solution report. Horizontal constraint was set to 0.01m and vertical set to 0.05m. These values are 10 times the estimated precision from Auspos.

A network of survey control was adjusted and reported for each island, forming the basis for the acquisition of ground test points. Full details are reported in the Geodetic Survey Report, presented with this report.

3.2 LiDAR Point Density Analysis:

For each Island 1km x 1km tiles were selected at random in order to test the point density of the data is within specification. The only specifications used in selecting which tile to test were the tiles had to be full data tiles which had only been covered once. In other words the tiles did not include any cross lines or re-runs. Please see the table below showing the results.

Area	Tile Number	Emitted Ground Point Density per sqm
Efate	e233n8051	6.358
	e227n8058	5.709
	e222n8058	5.973
	e216n8058	5.441
	e212n8056	5.662
	e209n8053	6.522
	e205n8048	6.299
	e205n8040	5.844
	e212n8043	6.117
	e214n8040	6.024
	e218n8037	6.606
	e217n8032	6.378
	e225n8032	6.065
	e224n8029	6.038
	e230n8031	6.008
	Average Point Density	6.070
Malekula	e749n8233	5.986
	e750n8230	7.025
	e754n8228	5.941
	e753n8225	6.665
	e756n8220	5.817
	e762n8215	6.147
	e764n8212	6.115
	e771n8206	6.985
	e778n8200	6.126
	e786n8196	5.203
	e789n8191	6.299
	e796n8186	6.292
	e800n8179	6.789
	e796n8175	6.025
	e800n8173	6.475
	Average Point Density	6.259

3.3 LiDAR Validation Analysis:

The LiDAR data has been checked against validation data on open clear land collected during the land survey. Comparison has been made with several validation areas across both of the Islands. The data was then adjusted and re-examined against the validation plots. Below are the tabulated results for each island showing the shifts applied and the comparison after the shifts.

3.3.1 Efate Topographic Analysis

Before Adjustment								
	Site							
	TIP1_MSL_121116	TIP2_MSL_121116	TIP3_MSL_121116	TIP4_MSL_121116	TIP5_MSL_121116	TIP6_MSL_121116	TIP7_MSL_121116	TIP8_MSL_121116
Average dz	0.583	0.504	0.576	0.651	0.639	0.724	0.678	0.616
Minimum dz	0.508	0.451	0.565	0.606	0.573	0.705	0.632	0.531
Maximum dz	0.673	0.558	0.592	0.725	0.670	0.767	0.862	0.787
Average magnitude	0.583	0.504	0.576	0.651	0.639	0.724	0.678	0.616
Root mean square	0.584	0.504	0.576	0.652	0.639	0.725	0.681	0.618
Std deviation	0.040	0.016	0.011	0.033	0.021	0.014	0.062	0.055
Average Dz Applied	-0.621							
After Adjustment								
	Site							
	TIP1_MSL_121116	TIP2_MSL_121116	TIP3_MSL_121116	TIP4_MSL_121116	TIP5_MSL_121116	TIP6_MSL_121116	TIP7_MSL_121116	TIP8_MSL_121116
Average dz	-0.041	-0.114	-0.044	0.031	0.019	0.104	0.058	-0.001
Minimum dz	-0.258	-0.169	-0.055	-0.014	-0.047	0.085	0.012	-0.089
Maximum dz	0.053	-0.030	-0.028	0.105	0.050	0.147	0.242	0.224
Average magnitude	0.047	0.114	0.044	0.034	0.025	0.104	0.058	0.047
Root mean square	0.063	0.116	0.045	0.045	0.028	0.105	0.083	0.061
Std deviation	0.048	0.019	0.011	0.033	0.021	0.014	0.062	0.061
Average Residual Dz	0.002							

3.3.2 Malekula Topographic Analysis

Before Adjustment				
	Site			
	TIP09_MSL_031212	TIP10_MSL_031212	TIP11_MSL_031212	TIP12_MSL_031212
Average dz	0.701	0.792	0.825	0.730
Minimum dz	0.632	0.720	0.717	0.681
Maximum dz	0.762	0.875	0.883	0.792
Average magnitude	0.701	0.792	0.825	0.730
Root mean square	0.702	0.793	0.826	0.730
Std deviation	0.030	0.040	0.036	0.024
Average Dz Applied	0.762			
After Adjustment				
	Site			
	TIP09_MSL_031212	TIP10_MSL_031212	TIP11_MSL_031212	TIP12_MSL_031212
Average dz	-0.066	0.032	0.068	-0.030
Minimum dz	-0.299	-0.040	-0.004	-0.079
Maximum dz	0.002	0.115	0.123	0.032
Average magnitude	0.066	0.040	0.068	0.033
Root mean square	0.082	0.050	0.075	0.039
Std deviation	0.050	0.040	0.032	0.024
Average Dz Applied	0.001			

3.3.3 Bathymetric Post Survey Accuracy Report

Please find below the Total Propagated Uncertainty Model for the flight configuration of Hawkeye IIB utilised on the Geoscience Australia Contract G4098 for the bathymetric LiDAR capture over Efate. The achieved accuracy of the soundings is within 0.55 meters (95% confidence) to depths of 30m.

Lidar Error Budget	A priori		Depth Range (m)				
	X/Y	Z	5	10	20	30	40
Tidal Measurements	0.05	0.05					
Co-Tidal model	0.00	0.00					
Base Station Position	0.05	0.05					
Navigation Solution GNSS + AV410	0.05	0.05					
HEII Internal Offsets and Latency	0.002	0.002					
HEII Lever Arms	0.005	0.005					
Scanner Position	0.001	0.001					
Sea Control Area	0.5	0.10					
Calibration Values	0.300	0.15					
Laser Discrimination	0	0.11					
Footprint Uncertainty x/y	2.200		2.256	2.417	2.973	3.720	4.565
Surface Angle of Incidence x/y	0		0.234	0.469	0.938	1.407	1.876
Surface Angle of Incidence z		0	0.063	0.125	0.251	0.376	0.501
Totals	2.278	0.244					
A posteriori x/y			2.344	2.531	3.173	4.021	4.971
A posteriori z			0.255	0.285	0.381	0.502	0.634
Order 1 Vertical Requirements			0.504	0.516	0.596	0.634	0.721

Provided by Applanix. Assumes a post-processed DGPS solution using a base station not more than 35km from the survey area

¹ Assumes a latency of 1ms and internal measurements of 2mm

¹ Uses output from POS PAC MMS iterated and verified by tape measure

¹ Assumes AHAB can record the angle to 0.001m

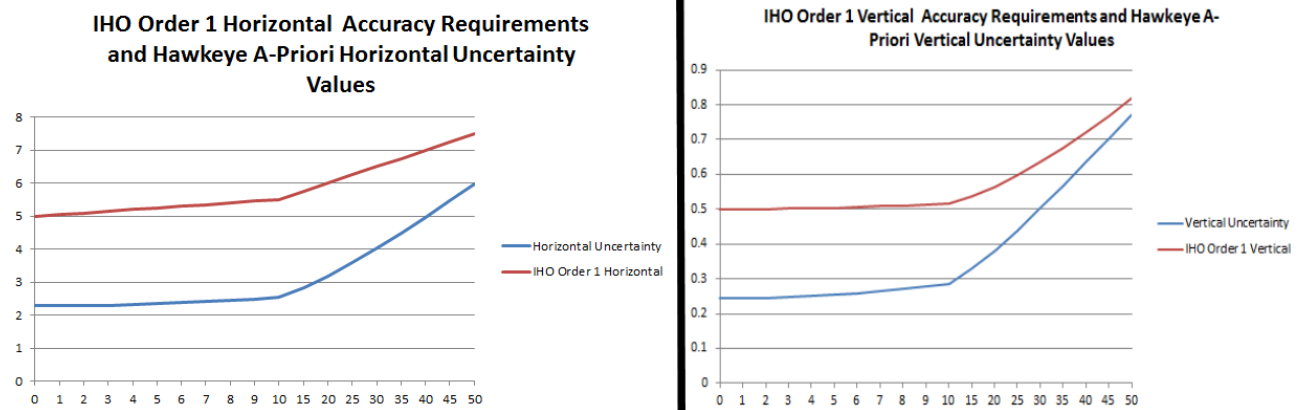
¹ Specification required by Pelydryn. May be better than this

¹ Discrepancy between the Control Area and the LiDAR data. This correction may already include errors from the Sea Control Area

¹ Based on the wavelength of light measured at 1ms pulse duration

¹ Spot Position Uncertainty in x/y is made up of the diameter of the footprint on the sea bed, divided by 2 for multi pixelation

¹ The angle of incidence at the sea surface will cause the light to enter the water at a different angle, rather than that assumed when the sea is flat (Snell's Law). Conditions during the survey were very good with sea state remaining around SS1. This is considered to create at worst a 10 degree error in the assumed angle of incidence at the sea surface. Spot Position Uncertainty in z is made up of the difference between the depth assuming the beam has entered the water when flat, and the depth assuming the beam entered the sea when it was at an angle of 10 degrees. This value is the diameter of the footprint divided by 4, which assumes the pixel position is in the middle of a quadrant of the footprint. 'c' is derived using pythagorus ($a^2 + b^2 = c^2$) where 'a' is the depth if the beam enters when the sea is flat, and 'b' is a quarter of the total footprint. This value is then calculated again for when the sea surface is angled at 10 degrees to the incoming laser beam to give a second value of 'c' (c2). The vertical error is the delta between 'c' and 'c2'.



4.0 Establishment of MSL from Tide Gauge Observations

4.1 MSL Introduction

In order to verify the accuracy of the EGM 2008 model or any adjustments to the vertical heights, the GeoScience Australia RFP directed that a number of tidal stations were to be recovered across the Vanuatu Island chain in order to establish any height differences between the 1'x1' EGM 2008 model and local Mean Sea Level. Following evaluation of the areas to be surveyed the following sites were identified:

- Main Wharf – Port Vila, Efate
- Emua Jetty, Efate
- Lits Lits Jetty, Malekula
- Luganville, Espiritu Santo

A tidal station was to be recovered/established at each site and sufficient observations were undertaken to allow several Mean Sea Level (MSL) observations to be made in order to establish any height differences. Any differences observed would be applied to the acquired topographic and bathymetric survey data in order to provide a seamless Digital Elevation Model.

4.2 Tidal Regime

Examination of existing tidal records revealed that the tidal regime is primarily semi-diurnal with a diurnal inequality evident. This results in a tidal regime that has two highs and two low waters a day, but one of these cycles will be a higher high and low water.

Tidal Ranges are generally small with an average range of around 1.0-1.5m. Despite there being little or no tidal information available for Malekula, it was assessed that the regime would be broadly similar across the areas to be surveyed.

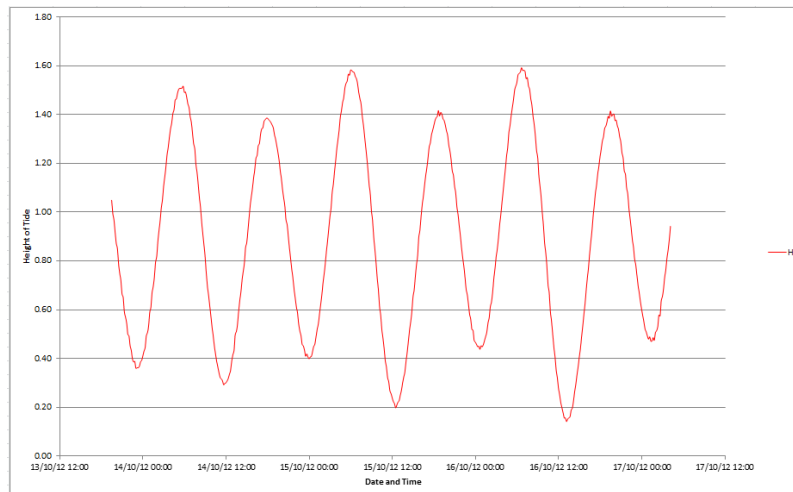


Figure 4.1 - Sample Tidal Curve Showing Tidal Regime

4.3 Recovery of Mean Sea Level

Surveyors have traditionally attempted to measure heights relative to MSL, which was thought to be a close approximation to the Geoid, and so it has become standard practice for authorities to evaluate MSL for a selected tide gauge and to relate the datum of all future levelling to this particular plane.

However a number of factors can cause variations in this level on a recurring cycle due to currents, water density changes, atmospheric pressure and winds. To determine a true mean sea-level, measurements must be made over long periods of time in order to average tidal variations. However, no matter how long a period of data is averaged, the true MSL is unattainable because changes are taking place over short and long time scales.

The results in this report are therefore a snapshot of levels based on an arithmetic average using the Low Pass Doodson Filter over 39 hour periods to remove diurnal and higher frequency tidal features.

The following methodology was used to recover/establish tide stations at each of the four locations identified:

- A GPS Benchmark was established adjacent to each tidal station using a minimum of 2 but ideally 6 hours of observations with a Dual Frequency GNSS Receiver. Static points were processed by AAM using the AUSPOS processing facility and the final co-ordinates were derived from a Network Adjustment of all points observed.
- For temporary tide stations a 2m tidepole and tidegauge sensor were co-located at a suitable point on the jetty with the zero of the pole set at a level below the lowest expected tide and as close to LAT as possible.
- The tidepole was levelled to the GPS Benchmark and in the case of Main Wharf in Port Vila to the local benchmark network.
- A 13 hour comparison was undertaken to confirm the correct operation of the gauge and to determine the relationship between the pole and gauge. Readings were taken at least every 30 minutes and at 10 minute intervals 1 hour either side of High and Low Waters.
- For the two temporary tide stations the gauges were left in-situ for periods in excess of 5 days.
- Regular 39hr Mean Sea Level Checks were undertaken to determine the value of MSL related to the ellipsoid.

- The averaged values for MSL at each site was compared with the EGM2008 value to establish the difference between the Geoid model and local Mean Sea Level to determine a revised orthometric height and correction to be applied to all data to adjust it from the ellipsoid to MSL.

4.3.1 Main Wharf – Port Vila

An existing gauge has been sited at this location (17° 45.409'S 168° 18.024E) since 1993 and is part of the South Pacific Sea Level and Climate Monitoring Project. In this instance the decision was made to recover the existing SEAFRAME gauge as it would provide a larger period of observations for checking MSL. A GPS Benchmark was established close to the existing SEAFRAME gauge and levelled to the Sensor mark (Van 16) and the reference mark (Van 14). These levelling runs were repeated and the results averaged to give an indication of the heights of all marks.



Figure 4.2 Location of Port Vila Tide Station

The tidepole installed at the station was damaged and was not usable. Due to the lack of a suitable location to insert a tidegauge in close proximity, a local mark was made levelled to the framework connecting the sensors to the jetty. Tidal observations were undertaken by measuring from this mark down to the water's surface. As the connection to Tide Gauge Zero was known it was possible to correct this reading to show the corrected Height of Tide. The observation was undertaken twice, first on 12 Oct 2012 and the second time on the 29th October 2012. This was due to an apparent lag in the time causing an offset on the data. Only the second observation was used to calculate the tide gauge zero and any corrections required.

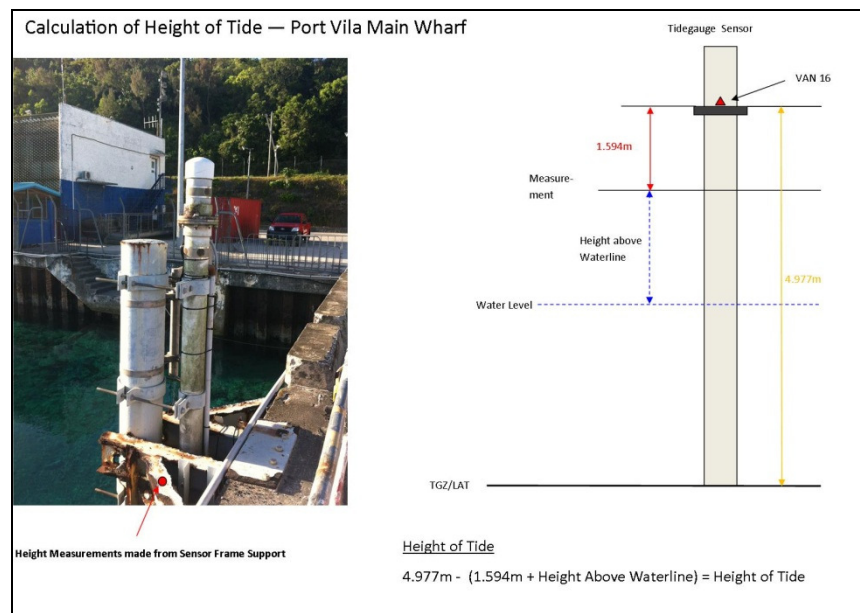


Figure 4.3 Port Vila Tide Station and Calculation of Height of Tide from Jetty

Following processing of the GPS and levelling observations the following connections were established:

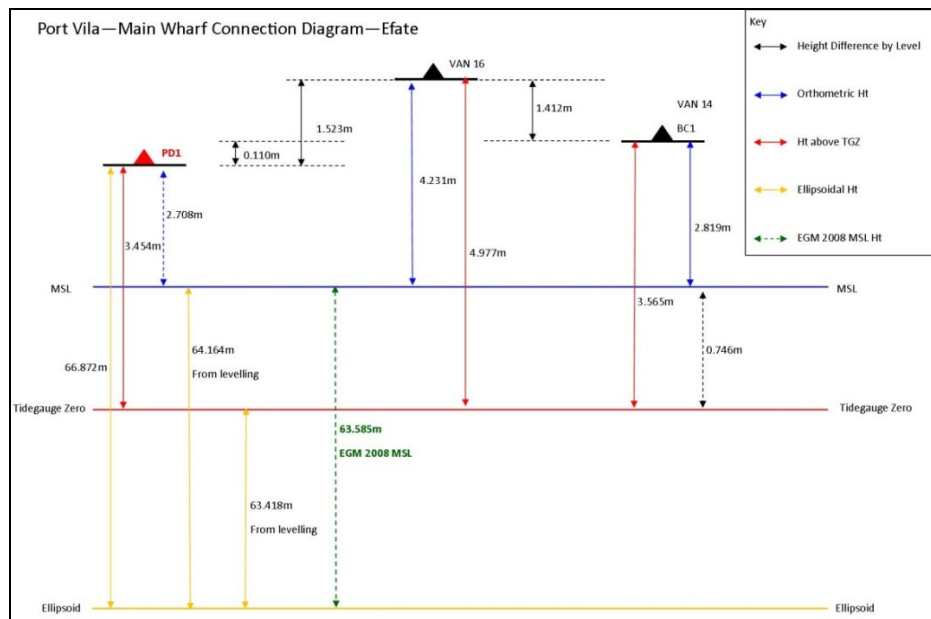


Figure 4.4 Port Vila Levelling Connection Diagram based on observations and quoted MSL values

Examination of the ellipsoid to geoid (MSL) values from levelling indicates a difference of 0.579m between EGM 2008 and the observed values. In order to validate this against the tidal record and the value of the Mean Sea Level that can be extracted the tidal data was analysed to confirm the tide gauge zero and derive MSL values. No harmonic analysis of the tidal data was undertaken. Following a 13 hour comparison the observed tidal readings were compared against the tidegauge using data downloaded from the NTF website to confirm the zero of the gauge with the following results:

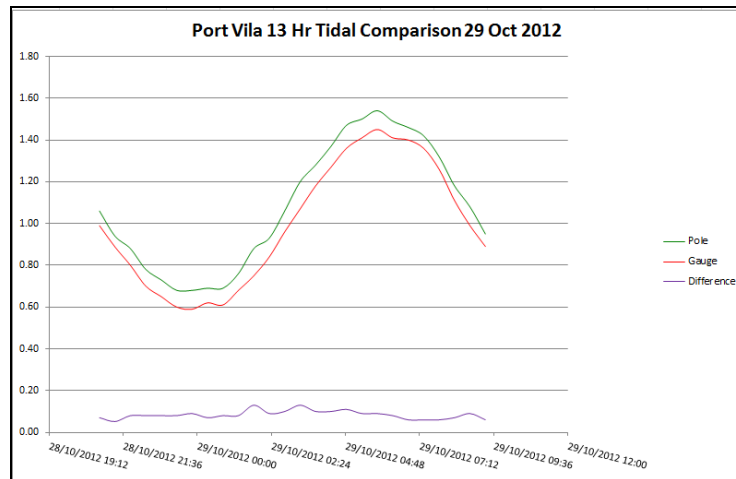


Figure 4.5 Graph showing difference between Tide Gauge and Observed Tidal Heights

From analysis of these readings it was inferred that 0.08m should be added to all gauge readings.

Having established the Tide Gauge Zero correction a series of Short Period Mean Sea Level Observations were undertaken using 39 hours of observations. The results from several observations (11, 21 and 28 Oct along with 15 Nov) were averaged (mean) and compared against the mean value of the 6 minute samples covering the period 02 Oct – 27 November 2012. From direct observation it was calculated that Mean Sea Level was 0.904m above the tidegauge zero.

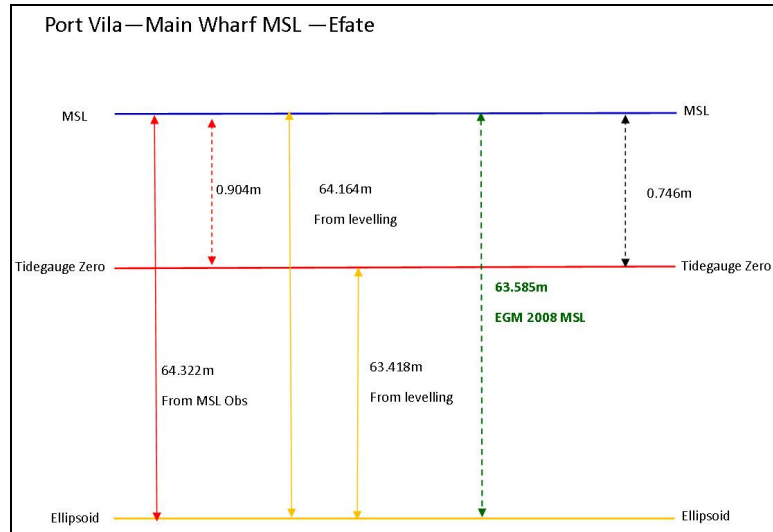


Figure 4.6 Observed, EGM 2008 and Calculated MSL Values Port Vila

Indications from the South Pacific Sea Level and Climate and Monitoring Project suggest that sea level in Vanuatu is rising at a rate of 5mm per year. Lowest Astronomical Tide was defined in 1973 at 0.746m. The observed MSL value of 0.904m is 0.158m different. This difference equates to the expected change in MSL since 1973 of 0.15m. This direct comparison gave confidence in the readings and the observed Mean Sea Level was accepted.

4.3.2 Emua Jetty - Efate

Following topographic surveying on the Northern side of Efate it was necessary to confirm the value of MSL in this locale. No tide station existed, however a substantial concrete jetty was

situated in an ideal location and permission was granted by the local chieftain and lease holder to establish a temporary tide station consisting of a 2m tidepole and Valeport 740 automatic tidegauge at Emua Jetty (17° 32.190' S 168° 23.167' E).

The Valeport Model 740 tide gauge has been designed to provide an accurate, easily deployed tide gauge for use in short or long term hydrographic survey operations. Low power consumption and user selectable sampling regime allow up to 2 years' autonomous operation. The transducer is vented to atmosphere and is IP 67 rated (0.5m for 30 seconds)

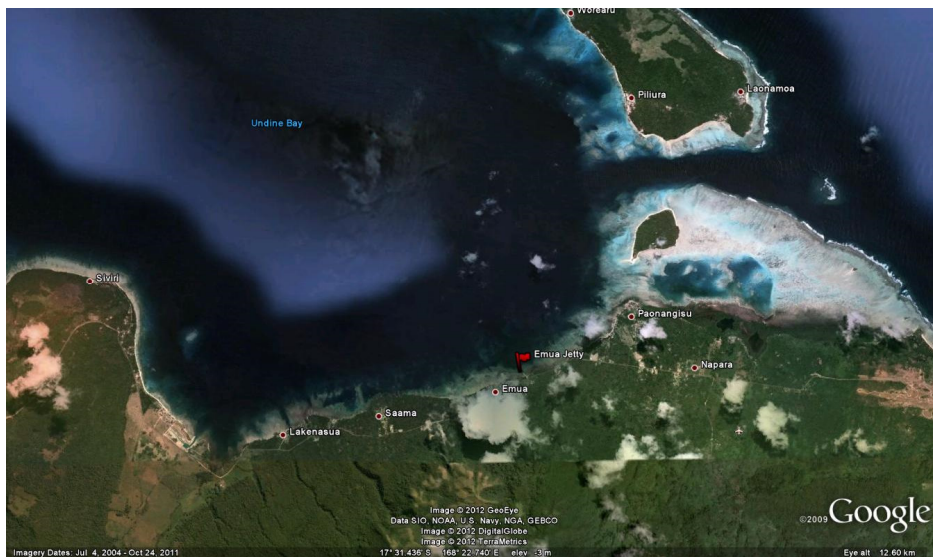


Figure 4.7 Location of Emua Tide Station

The jetty had a number of wooden rubbing strakes and the decision was made to fix the temporary tidal station to this with the recording unit chained to a bollard in a waterproof box. The pole and gauge were set at a level below Lowest Astronomical Tide using predicted tides to set the depth of immersion at installation. Following installation of the gauge a GPS Benchmark was established at the root of the jetty with observations taking place for 6 hours. The top of the tidepole was levelled to the new benchmark and this was used as the basis of deriving a Mean Sea Level.

The zero of the tidepole and the tidegauge sensor were fixed in an identical position on a board fastened to the rubbing strake. A 13 hour comparison was conducted between the gauge and pole and it was found that the zero of the gauge was coincident with tidepole zero.

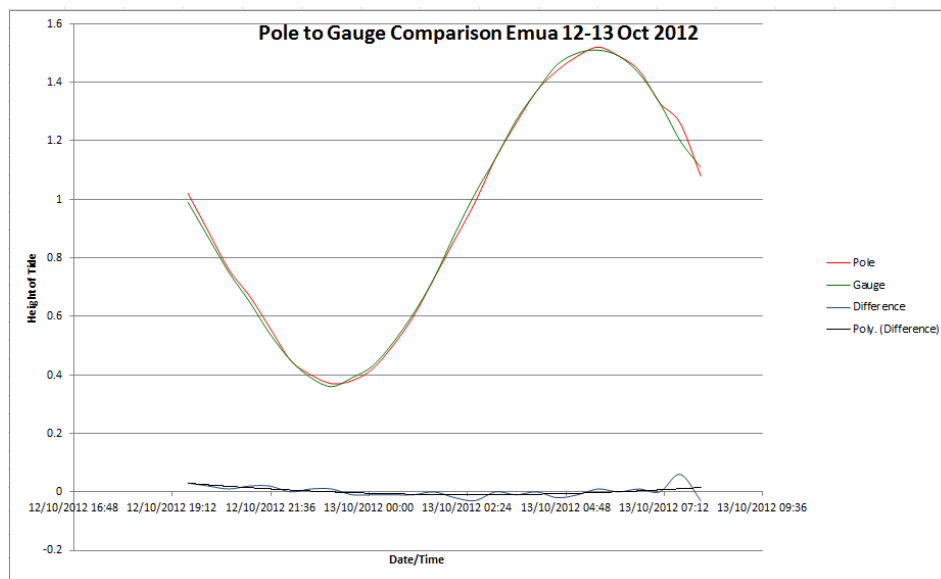


Figure 4.8 Graph showing difference between Tide Gauge and Observed Tidal Heights

Following processing of the GPS Benchmark the following connections and heights were established:

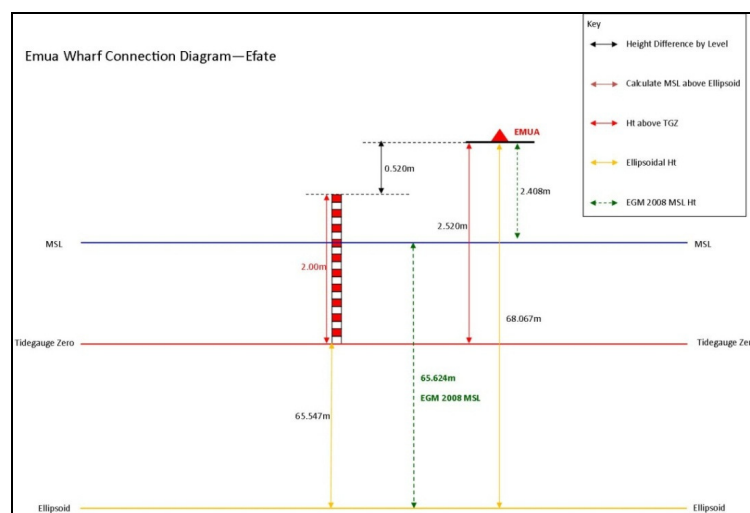


Figure 4.9 Emua Levelling Connection Diagram based on observations and EGM 2008 MSL values

The gauge was left in-situ for several days and then downloaded to allow Mean Sea Level calculations to be conducted. Several observations were made directly by reducing the tidal readings to the ellipsoid as the height of gauge zero was known (65.547m).

Having averaged the values of the MSL checks the ellipsoid-geoid separation value was calculated as 66.425m. Mean Sea Level was found to be 0.880m above the zero of the pole. This value was 0.801m higher than the EGM 2008 model, but shows broad agreement with the station at Port Vila (0.90m) and was accepted.

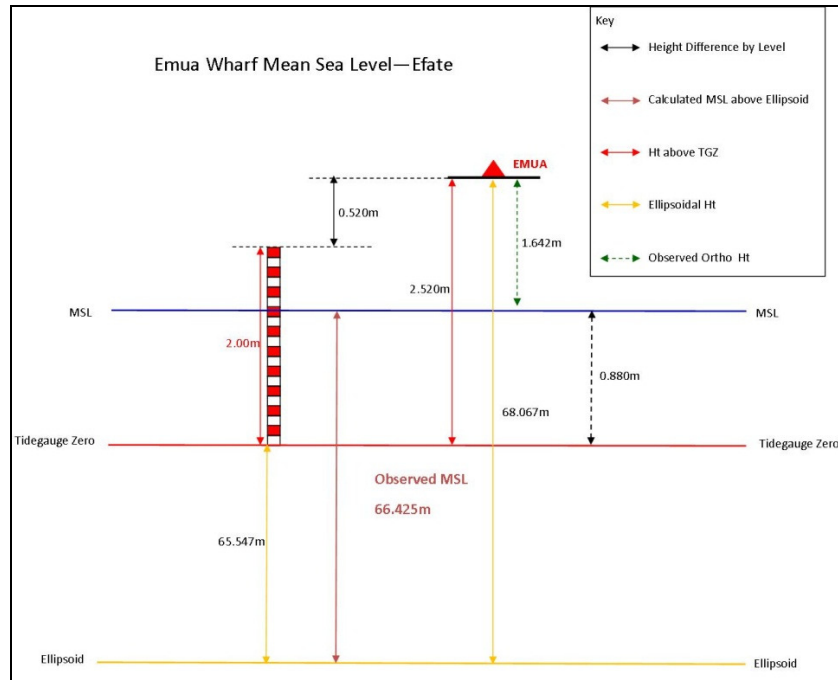


Figure 4.10 Observed, MSL Values Emua

The decision was made to leave the station in-situ as the programme had slipped due to aircraft availability and as a result it may have been possible to collect a full 28 days of observations. However on visiting the site on 20 October 2012, it was found that the station had been vandalised and the station damaged beyond repair. The loss of this and any additional data was reported separately. It is considered the values obtained give a good indication of the Mean Sea Level on this side of the island.

4.3.3 Lits Lits Jetty - Malekula

Part of the overall contract involved topographic LiDAR surveys along parts of the island of Malekula, which lay to the North of Efate. In order to reduce heights to Mean Sea Level it was therefore necessary to confirm the value of Mean Sea Level at some point on the island of Malekula. No tide stations existed on the island, however a substantial concrete jetty was situated in an ideal location midway along the east side of the island at Lits Lits Jetty (16° 06.750' S 167° 26.622' E)

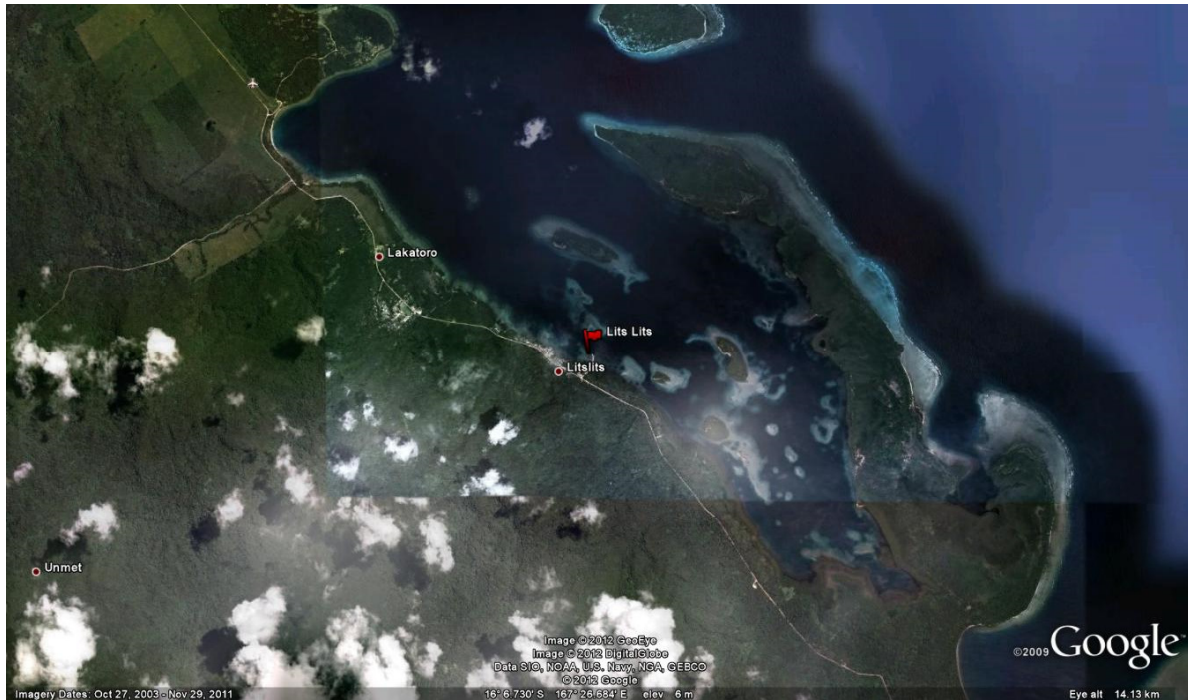


Figure 4.11 Location of Lits Lits Tide Station

Permission was granted by the local chieftain and lease holder to establish a temporary tide station consisting of a 2m tidepole and Valeport 740 automatic tidegauge.



Figure 4.12 Malekula Jetty and Tide Station

The jetty had a number of large rubber rubbing strakes and the decision was made to fix the tidal station to this with the recording unit chained to a bollard in a waterproof box. The pole and gauge were set at a level below the expected LAT using predicted tides. Following installation of the gauge a GPS Benchmark was established at the opposite side of the jetty with observations taking place for 6 hours. The top of the tidepole was levelled to the new benchmark and this was used as the basis of deriving a Mean Sea Level.

A 13 hour comparison was conducted between the gauge and pole and it was found that the zero of the gauge was 0.02m shoaler than Tide Pole Zero and 0.02m should be added to gauge all readings.

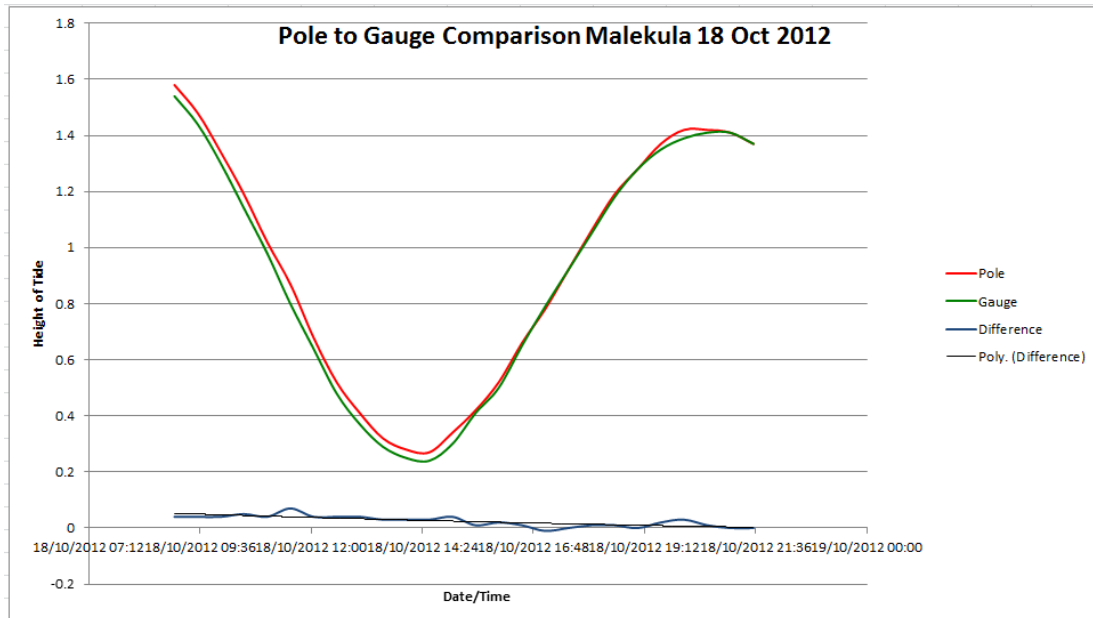


Figure 4.13 Graph showing difference between Tide Gauge and Observed Tidal Heights

Following processing of the GPS benchmark the following connections were established:

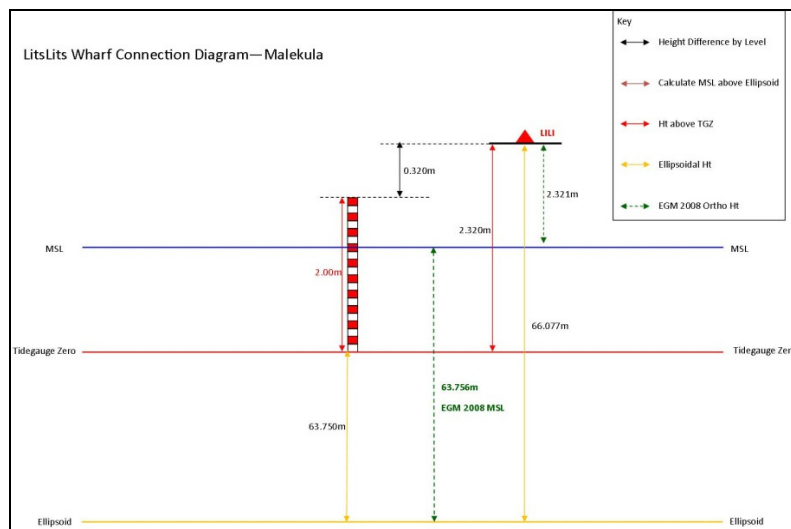


Figure 4.14 Malekula Levelling Connection Diagram based on observations and EGM 2008 MSL values

The gauge was left in-situ between 18th October and 25th October 2012 and then the gauge was recovered and downloaded to allow Mean Sea Level calculations to be conducted. Several observations were made directly reducing the tidal readings to the ellipsoid.

Having averaged the values of the MSL checks the ellipsoid-geoid separation value was calculated as 64.783m. Mean Sea Level was 1.033m above the zero of the pole.

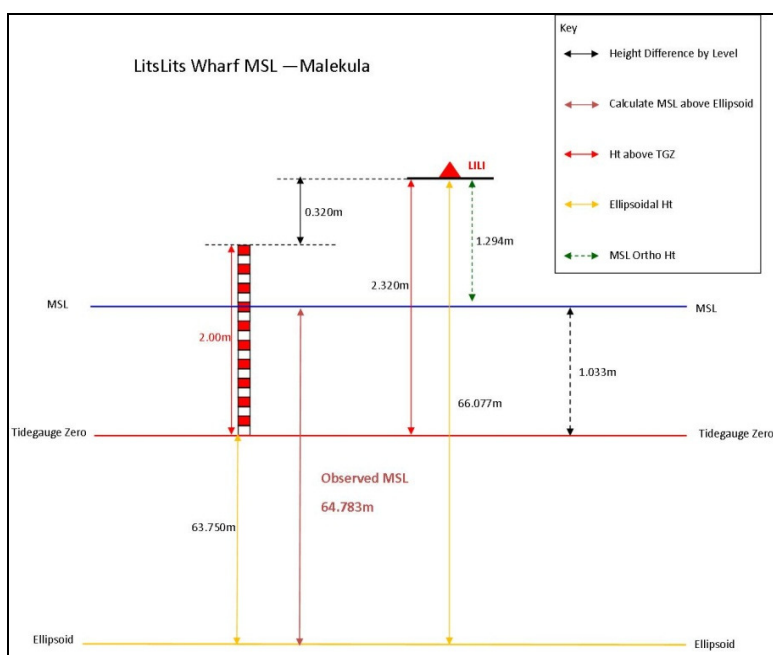


Figure 4.15 Observed, MSL Values Malekula

The derived Mean Sea Level appears to be consistent when compared to the values obtained in Efate and were accepted. The change in MSL is inferred to be a result of a change in the tidal regime and also possibly some impounding effects as the jetty was located in a lagoon area.

4.4 Mean Sea Level Comparisons

4.4.1 Comparison with EGM2008 Values

During processing of all of the GPS Benchmarks, orthometric heights were calculated using the 1'x1' EGM 2008 mark. As a result it was possible to calculate the ellipsoid to geoid (MSL) value for these points. These points were then compared with the derived values and the differences established as can be seen in the table below:

Station	Ellipsoidal Height	EGM 2008 Geoid-Ellipsoid Separation Value	Calculated Separation Value based on MSL	Difference between EGM 2008 and Observed MSL Separation Values
PoVi	66.872	63.585	64.322	-0.737
Emua	68.067	65.624	66.425	-0.801
LiLi	66.077	63.756	64.783	-1.027

4.4.1 Comparison with Existing MSL Data

In order to validate the Mean Sea Level Values calculated through the use of Short Period Observations a comparison was made of the readings at Port Vila against longer period observations held by the South Pacific Sea Level Climate Monitoring Project (SCSLCMP). As all of the other island Mean Sea Level values were in close agreement with Port Vila Mean Sea Level, it was felt that this comparison validates all of the results. Several epochs of data were downloaded to allow averages to be calculated from yearly, monthly and year to date values with the following results.

Annual MSL Values	Level (m)	Sea Level (m)	
2010	0.899	0.904	+0.005
2011	0.919		-0.015
2012 (to Oct)	0.905		+0.001
2010-2012	0.908		-0.004

A further comparison was done against the monthly values of 2012 (to Oct) with the following results.

Comparison with 2012 Monthly MSL Values	Mean Sea Level (m)	Observed Mean Sea Level (m)	Difference
January	0.952	0.904	-0.048
February	0.883		+0.021
March	0.909		-0.005
April	0.853		+0.051
May	0.878		+0.026
June	0.916		-0.012
July	0.899		+0.005
August	0.933		-0.029
September	0.940		-0.036
October	0.890		+0.014
Sum of 2012	0.906		-0.002

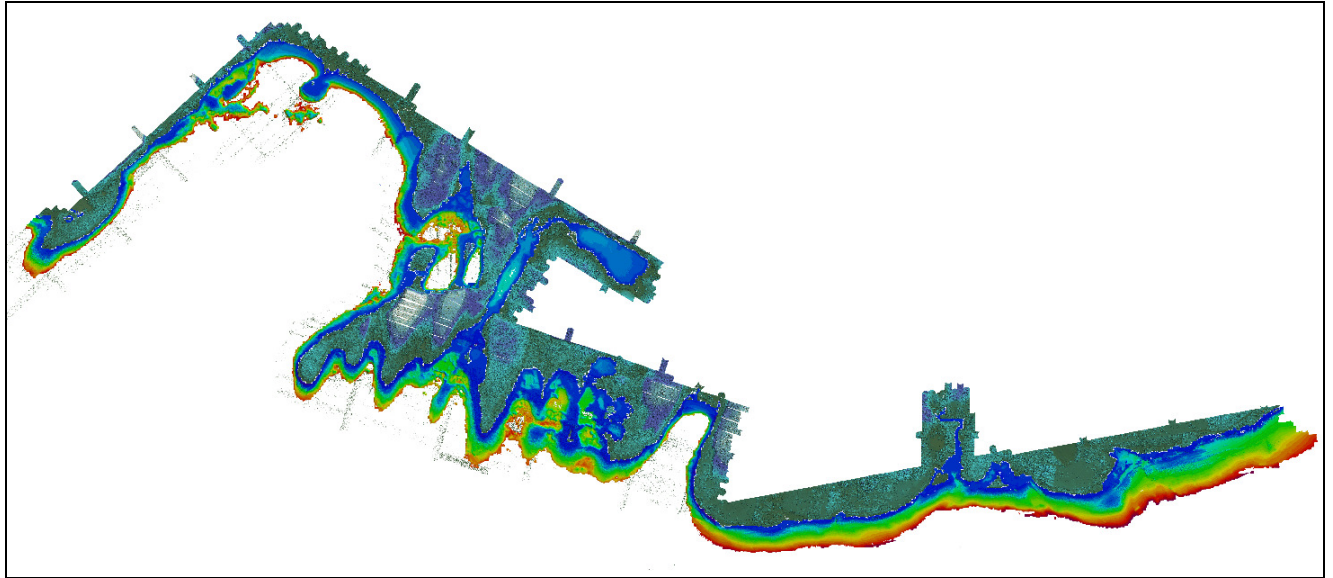
Analysis of this data sees a larger variation in the values on a month by month basis and this is accepted to be a result of seasonal variations in climate. However the values are consistent and show that the calculated Mean Sea Level has been established correctly.

It is assessed that following analysis of the data that the Mean Sea Level Values obtained are accurate to 0.004m when compared to the last 3 years of tidal records at Port Vila. These differences seen are within the likely range of accuracies as stated in The International Oceanographic Commissions Manual on Sea Level Measurement and Interpretation and it is recommended that the values derived are used to reduce ellipsoidal data to MSL.

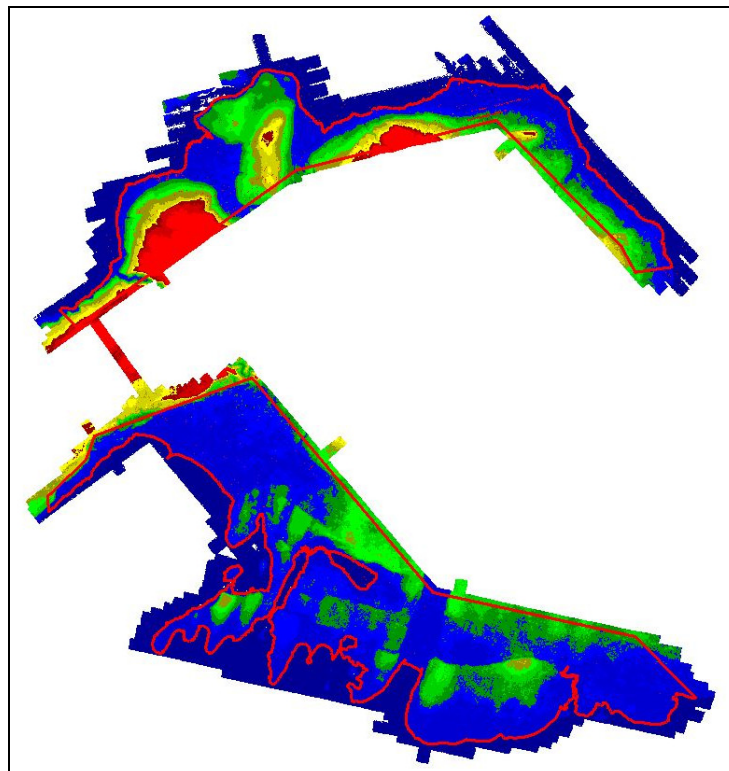
The results obtained are a good indicator of the differences between the EGM 2008 and local orthometric height datum and should be used to correct the data. It is assessed that the MSL values are accurate to +/- 0.004m with only average differences of 0.05m seen during the period of observations and analysis. The insertion of temporary GPS benchmarks at Luganville will allow a local connection to be derived and allow the data from the gauge to be more fully analysed.

5.0 Validation Plots for LiDAR and Imagery

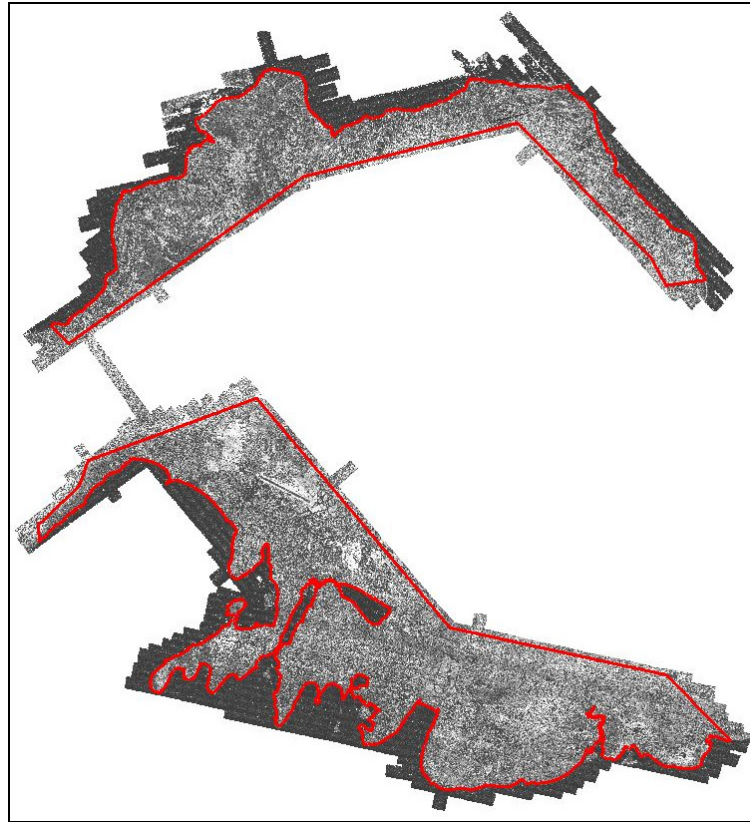
5.1 Efate Bathymetric LiDAR Capture



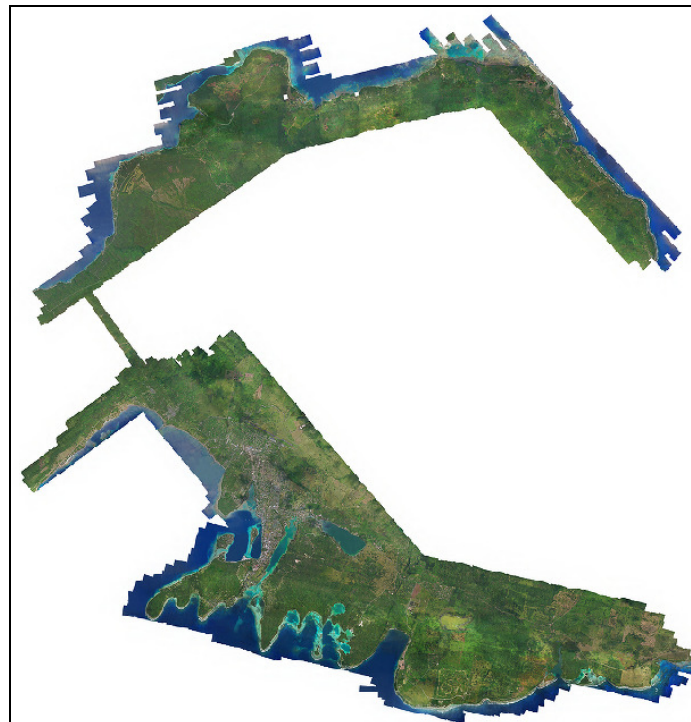
5.2 Efate Topographic Elevation



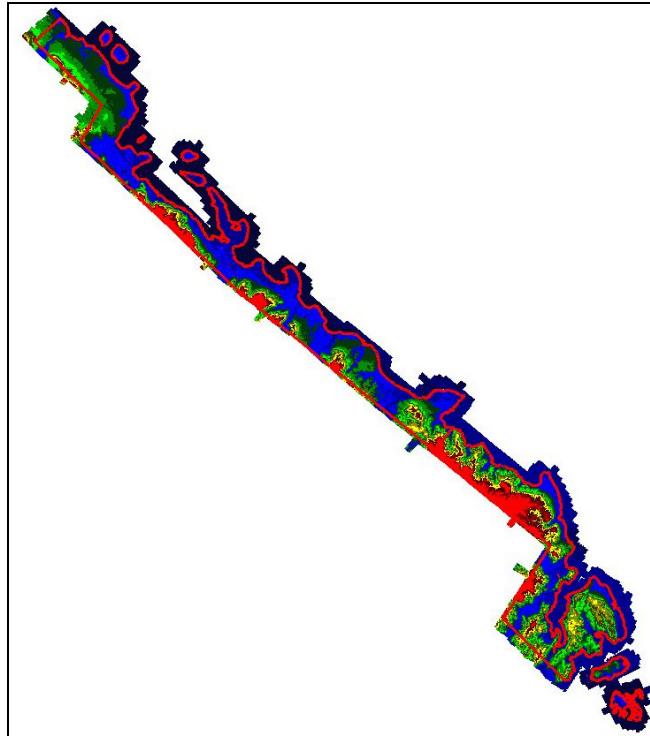
5.3 Efate Topographic Intensity



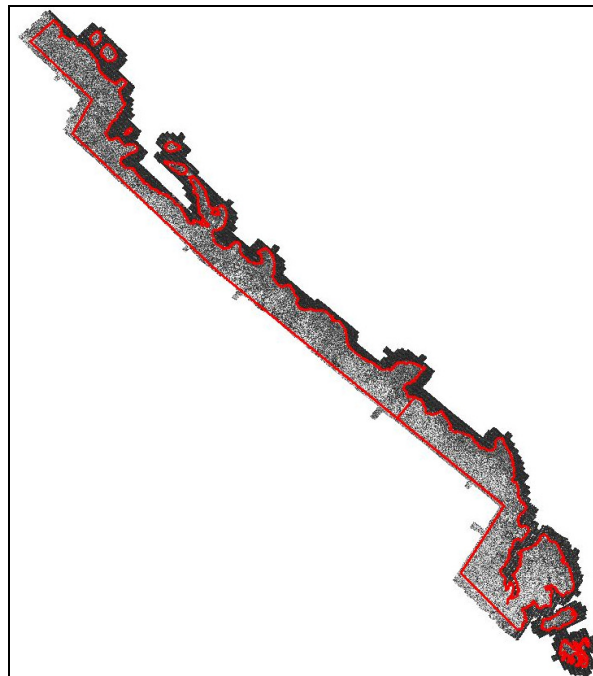
5.4 Efate Topographic Imagery



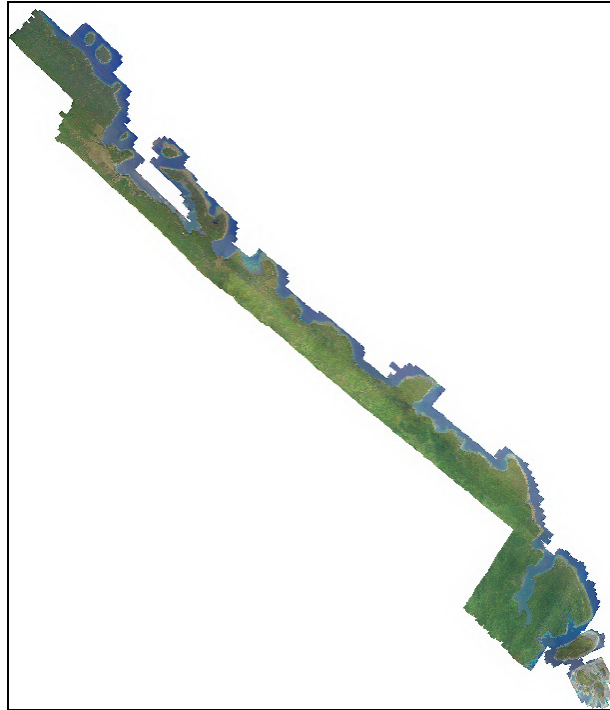
5.5 Malekula Topographic Elevation



5.6 Malekula Topographic Intensity



5.7 Malekula Topographic Imagery



6.3 Geodetic Survey Report

1. INTRODUCTION

AAM were engaged by Geoscience Australia (GA) to deliver high resolution airborne Topographic and Bathymetric LiDAR and derived products for selected areas across the islands of Efate, Espiritu Santo and Malekula, Vanuatu.



Figure 1 – Vanuatu

2. BACKGROUND INFORMATION

Extracted from the Geoscience Australia Project Brief:

“As Part of the Australian Government’s International Climate Change Adaptation Initiative (ICCAI), the Pacific Adaptation Strategy Assistance Program (PASAP) aims to assist in the development of evidence based adaptation strategies to inform robust long-term national planning and decision making in partner countries.

The primary objective of PASAP is to enhance the capacity of partner countries to assess key vulnerabilities and risks, formulate adaptation strategies and plan, in addition to mainstreaming adaptation into decision making.

The Government of Vanuatu has identified coastal inundation of infrastructure and settlements as one of the highest national adaptation priorities, reflecting the concentration of the population and infrastructure along the coast. In order to support vulnerability assessments and adaptation planning, high resolution digital elevation models (DEM) and mapping were also identified as an immediate need.

With the assistance of Geoscience Australia a number of areas were identified for capture by Topographic and Bathymetric LiDAR. To support the delivery of accurate information, ground survey was required to compile a control network and measured ground test points.”

3. SCOPE OF SURVEY

In planning the capture of the two types of LiDAR datasets over the areas of interest provided by Geoscience Australia, AAM determined nineteen (19) test point areas (TIP) were required across the 3 islands to validate and QA the LiDAR data.

Eight (8) specific sites of the nineteen (19) sites were to be located close (~200m) to the shore to test the Topographic and Bathymetric LiDAR data sets. The remaining eleven (11) sites were to test the Topographic LiDAR data only.

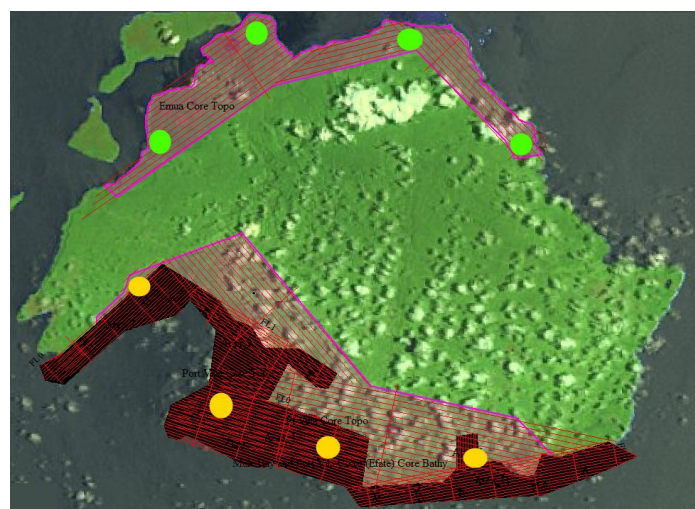
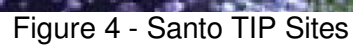
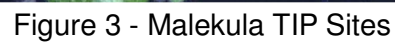


Figure 2 - Efate TIP Sites



The scope of survey work to acquire and report test point data included; survey control, test point survey and combined processing of data with Pelydryn to determine MSL heights.

3.1 Control Network – GPS

A breakdown of the GPS survey includes:

- At each test point site (TIP), place at least two new control points in open space suitable for static GPS observations, that has a variety of vegetation types able to be measured by total station:
 - Clear ground
 - Grassland
 - Low lying bushes and shrub land
 - Woodland and open forest
 - Dense Vegetation
- Proposed TIP sites could be moved to areas with suitable vegetation types
- For each island create one Primary Control Network with connecting static GPS measurements to:
 - All new control points at TIP sites
 - Any permanent survey control as supplied by the Land Survey Department
 - The airport base station (primary and secondary)
 - The tide gauge survey mark
- Where no existing marks or control values exist, measure long static sessions of data
- Resolve long static observations using Auspos (Geoscience Australia)
- Process and adjust network using Auspos solution holding fixed the airport base station control values

3.2 Test Points

The requirement for test point data was to measure 20-25 points under and adjacent various vegetation types. With the thick vegetation, conventional survey methods were used. A Leica TS15 Total Station was used to acquire the coordinate data (E, N, H).

At the time of test point survey control coordinates were not available. Data was collected in an arbitrary grid/datum and corrected post survey using a CAD package.

3.3 Processing - Office

The processing of survey data included numerous stages subject to receiving the final Auspos coordinates and the tide gauge resolved MSL heights from Pelydryn after the field survey was completed.

After applying the correct final horizontal coordinates and MSL height values, the survey included the list of test point coordinates at each TIP site in the correct grid and datum:

<u>Island</u>	<u>Horizontal Grid</u>	<u>Vertical Datum</u>
Efate	UTM 59	MSL
Malekula, Santo	UTM 58	MSL

Figure 1 – Grids and Datums

Note: MSL is GPS resolved orthometric heights using geoid model EGM2008 with a vertical shift applied onto the known MSL value at each tide gauge control point. Refer (D) Table 3.

The deliverables include; TIP coordinate listing, photos, and report.

4. GRIDS AND DATUMS

Currently there is only one valid horizontal grid and vertical datum in use across Vanuatu.

4.1 Horizontal Grid Parameters

Projection	Universal Transverse Mercator (South)
False Easting	500,000.000 m
False Northing	10,000,000.000 m
Central Meridian	165 (Zone 58) Santo & Malekula
	171 (Zone 59) Efate
Central Scale Factor	0.9996
Latitude of Origin	0
Reference Ellipsoid	GRS80
Semi-Major Axis	6378137.0000
Inverse Flattening	298.25722210088

Table 2 – Projection Parameters

4.2 Vertical Datums

The vertical datum for the survey is Mean Sea Level. MSL was derived by a combination of methods:

1. GPS computation using the geoid model EGM2008
2. Tide gauge observations at wharf locations on each island

From a comparison of these measurements it was deduced EGM2008 has approximately one (1) metre of error in the interpolated geoid separation.

The shape of the geoid model remained unchanged, just the geoid separations were modified by approximately 1 metre to produced the tide resolved MSL heights.

MSL heights on wharf survey marks							
Wharf Control Point (tide gauge connected)		Phase 1 Resolved Heights (GPS only)			Phase 2 Adopted Heights (Tide gauge)		
		<u>Ellipsoid Height</u> (GRS80)	<u>Geoid Separation</u> (EGM2008)	<u>Orthometric height</u> (EGM2008)	<u>MSL</u> (measured)	<u>Ellipsoid Height</u> (GRS80)	<u>Geoid Separation</u> (corrected)
<u>Island</u>	<u>ID</u>	(A)	(B)	(C)	(D)	(E)	(F)
Efate	povi	66.872	63.585	3.287	2.550	66.872	64.322
	emua	68.067	65.624	2.443	1.642	68.067	66.425
Malekula	lili	66.077	63.756	2.321	1.294	66.077	64.783
Santo	saw1	67.635	64.855	2.780	1.750	67.635	65.885
	saw2	67.573	64.854	2.719	1.688	67.573	65.885

Table 3 – Mean Sea Level Calculations

5. CONTROL SURVEY

5.1 Existing Control

The Vanuatu Land Survey Department provided a number of control stations along the coastline adjacent to the mapping polygon that AAM would use where found.

A handheld GPS was used to locate existing control points from the geographic (Lat/Long) coordinates provided by the Vanuatu Land Survey Department (Tony Kanas). The existing control points found:

- Efate Z13, 223, and Z8
- Malekula Lemap, Norsup
- Santo HOGB



Figure 5 – Z13



Figure 6 - 223



Figure 7 – Z8



Figure 8 - Lemap



Figure 9 – Norsup



Figure 10 - HOGB

Other permanent survey control marks provided by the Vanuatu Land Survey Department were found to be unuseable. Local knowledge was essential to locate these marks as most had no witness marking.

5.2 Airport Base Stations

At the Port Vila (Efate) and Luganville (Santo) International Airports, AAM established Primary and Secondary survey control points to observe base GPS data for the LiDAR capture. These survey control points are permanent structures for future reference and were connected as part of ground survey GPS network.



Figure 11 - Airport Base GPS (Efate)

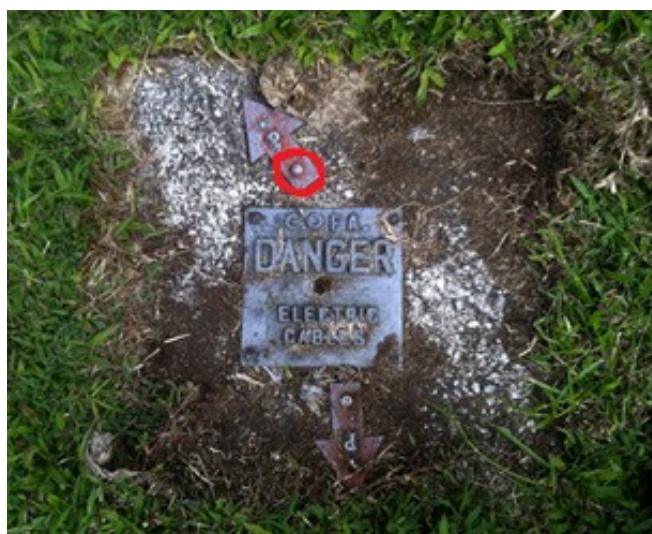


Figure 12 – Primary Airport Station (Efate)



Figure 13 – Secondary Airport Station (Efate)



Figure 14 - Airport Base GPS (Santo)



Figure 15 - Primary Station (Santo)

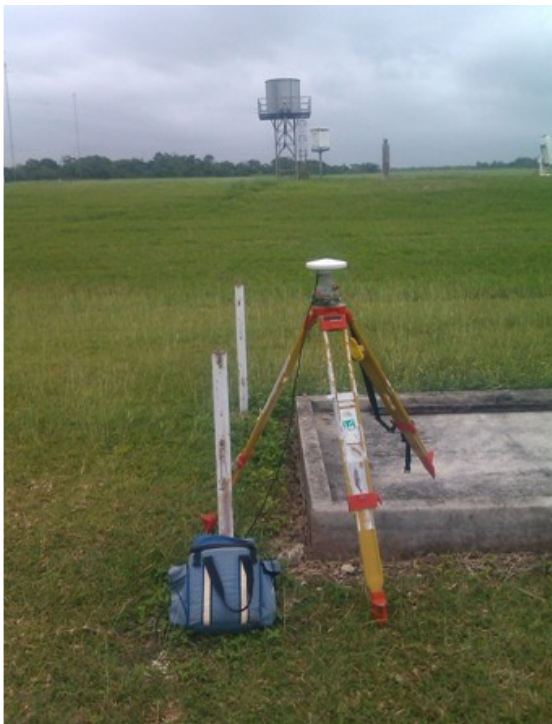


Figure 16 – Secondary Airport Station (Santo)

6. NEW SURVEY CONTROL

After locating the existing survey control provided by the Vanuatu Land Survey Department, new survey control marks were required at the test point locations (TIP) to allow for ground survey measurements of approximately 100 points.

In all nineteen (19) sites at least two points were placed and measured into the network using GPS. These were only temporary marks.

6.1 Wharf Survey Stations

At four (4) wharf locations across the three islands permanent survey control marks were installed. These marks were used to:

- Complete a long static observation for Auspos solution
- Connect by static GPS into the network
- Complete a level connection to the tide gauge

The fifth wharf mark at the main wharf in Villa was an existing mark.

All marks placed were stainless steel pins or similar into the concrete or bitumen surface near to the tide gauge. For detailed information regarding these stations refer to the Post Survey Spatial Accuracy Report.



Figure 17 - Typical Wharf Station

6.2 Primary Control Station Coordinates

The coordinates for the permanent survey control points provided by the Vanuatu Land Survey Department were of unknown history and quality. It was decided all primary control marks (permanent) needed to be measured by long static observations and processed using Geoscience Australia Auspos solution.

In most situations 6 hours of data was observed to generate the primary control coordinates. This process meant that on each island there were a number of primary control marks including the airport control stations (Efate, Santo) to provide primary coordinates for the network solution.

6.3 Static GPS Network

Each island network consists of static baselines, connecting the adopted Primary Control Marks, airport control, and control marks at each TIP site. Each network was adjusted using Leica Geo Office software.

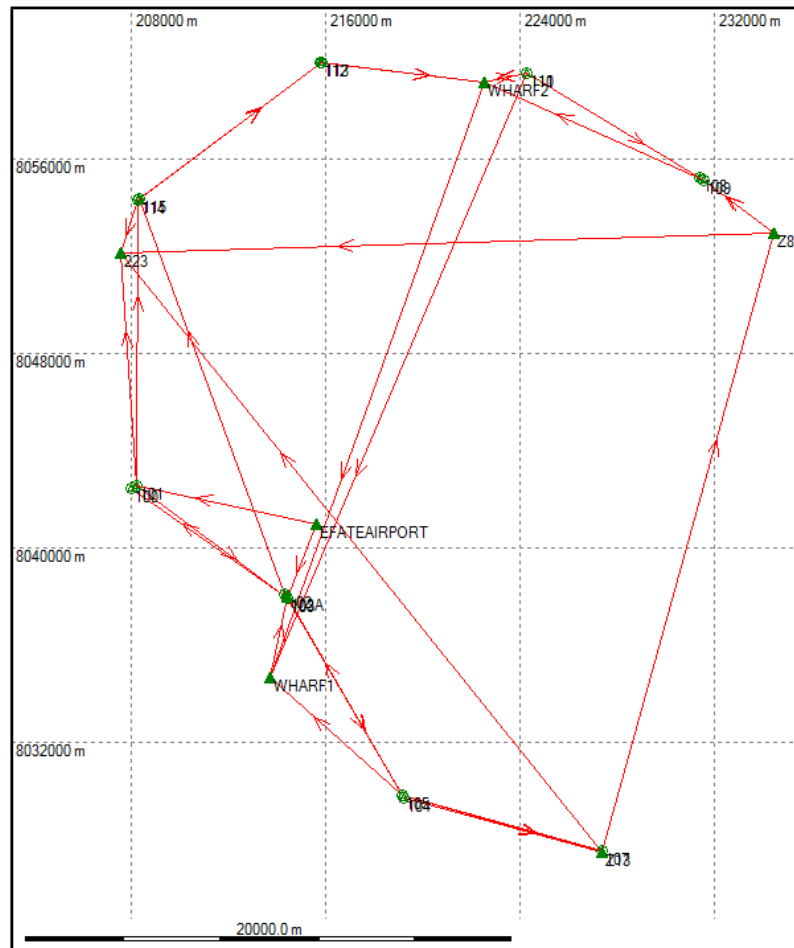


Figure 18 - Efate GPS Network

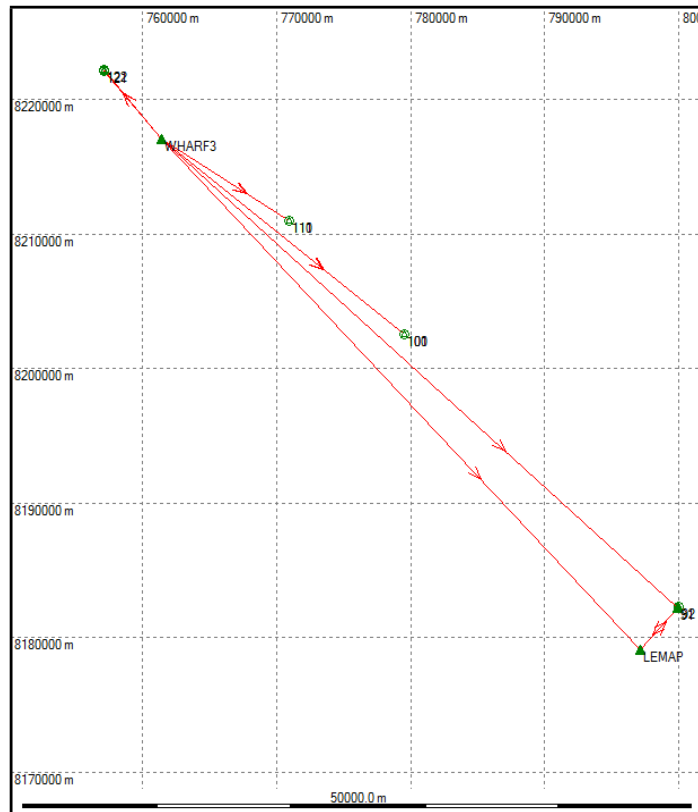


Figure 19 – Malekula GPS Network

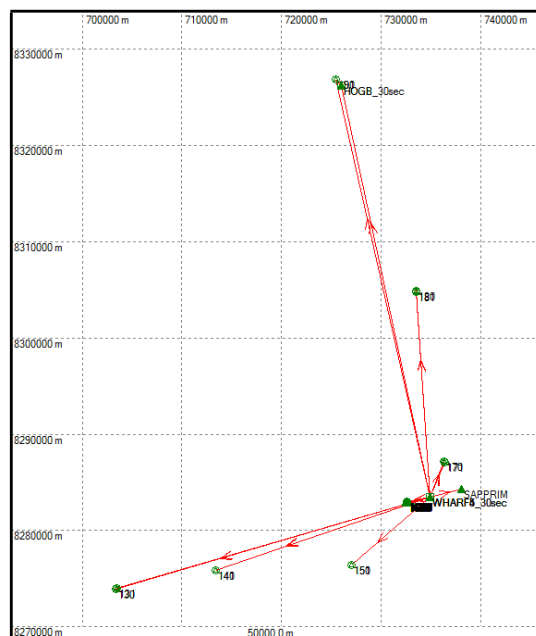


Figure 20– Santo GPS Network

6.4 Network Adjustment

The Primary control station coordinates were assigned as fixed points and constrained horizontally and vertically according to the Auspos solution report. These constraints were set to approximately 0.01m horizontally and 0.05m vertically. These values are 10 times the estimated precision from Auspos.

6.5 Survey Control Coordinates

A network of survey control was adjusted and reported for each island, forming the basis for the acquisition and processing of ground test points.

 SURVEY CONTROL - VANUATU LIDAR UTM Zone 50										
Job No: 20076A		Job Name: VANUATU LIDAR				Prepared by: ST				
Date: 7/12/12		Location: EFATE				Revision: 0				
AAM: AAM Group Pty Ltd VLS Vanuatu Land Survey PEL Pelydryn Pty Ltd VG Vanuatu Government				GPS: Static GPS TS: Total Station SL: Spirit Leveling TG: Tide Gauge		BP: Brass Plaque SHN: Spring head Nail SP: Steel Pin GP: Ground Plate CONC: Concrete				
Station	ITRF2008 Coordinates		Height	Description	Location	Ground Mark	Installed By	Method (H/V)	Estimated	
	Easting	Northing	TG						H	V
South Side of Efate										
WHARF1 (povi)	213718.036	8034663.940	2.550	Pin in concrete wharf (permanent) +/-0.04m	Main wharf	SP / CONC		VG	GPS	20ppm
VILA PRIMARY BASE (airport)	215602.397	8040986.833	17.594	Markings on steel plate +/-0.04m	airport	GP		VG	GPS	20ppm
100 +/-0.04m	208019.911	8042460.796	7.791	Nail (non permanent)	TIP1		SHN	AAM	GPS	20ppm
101 +/-0.04m	208235.317	8042568.954	3.126	Nail (non permanent)			SHN	AAM	GPS	20ppm
102 +/-0.04m	214338.782	8038079.977	1.463	Nail (non permanent)			SHN	AAM	GPS	20ppm
103 (removed)	214442.467	8037953.128	1.304	Nail (non permanent) +/-0.04m		TIP2	SHN	AAM	GPS	20ppm
103A +/-0.04m	214443.381	8037954.329	1.349	Nail (non permanent)			SHN	AAM	GPS	20ppm
104 +/-0.04m	219207.143	8029721.781	5.866	Nail (non permanent)	TIP3		SHN	AAM	GPS	20ppm
105 +/-0.04m	219182.408	8029836.343	5.131	Nail (non permanent)			SHN	AAM	GPS	20ppm
Z13 +/-0.04m	227372.033	8027477.549	2.825	Metal tube set in conc (permanent)			SP / CONC	VLS	GPS	20ppm
107 +/-0.04m	227402.904	8027534.935	6.304	Nail (non permanent)	TIP4		SHN	AAM	GPS	20ppm
North side of Efate										
WHARF2 (emua) +/-0.04m	222519.661	8059165.356	1.642	Pin in concrete wharf (permanent)			SP	PEL	GPS	20ppm

Table 4 - Efate Survey Control



RFQ NUMBER: RFTGA 2012 / 2863
ACQUISITION OF LiDAR, IMAGERY AND DERIVED PRODUCTS FOR
VANUATU
FINAL REPORT - MALEKULA



SURVEY CONTROL - VANUATU LIDAR

UTM Zone 58

Job No:	20076A	Job Name:	VANUATU LIDAR			Prepared by	ST			
Date:	7/12/12	Location:	MALEKULA			Revision	0			
AAM: AAM Group Pty Ltd VLS Vanuatu Land Survey PEL Pelydryn Pty Ltd VG Vanuatu Government			GPS: Static GPS TS: Total Station SL: Sprirt Leveling TG: Tide Gauge			BP: Brass Plaque SHN: Spring head Nail SP: Steel Pin GP: Ground Plate CONC: Concrete				
Station	ITRF2008 Coordinates		Height	Description	Location	Ground Mark	Installed By	Method	Estimated Accuracy	
	Easting	Northing	TG (EGM2008)					(H/V)	H	V
WHARF3 (Illi)	761373.858	8217066.306	1.294	Pin in concrete wharf (permanent)		SP / CONC	PEL	GPS	20ppm	+/-0.04m
LEMAP	797119.842	8179126.596	5.986	Pin in concrete (permanent)		SP / CONC	VLS	GPS	20ppm	+/-0.04m
91	799932.606	8182262.723	25.088	Nail (non permanent)	TIP9	SHN	AAM	GPS	20ppm	+/-0.04m
92	800015.839	8182302.301	24.174	Nail (non permanent)		SHN	AAM	GPS	20ppm	+/-0.04m
100	779534.526	8202516.310	34.593	Nail (non permanent)	TIP10	SHN	AAM	GPS	20ppm	+/-0.04m
101	779491.241	8202538.972	36.356	Nail (non permanent)		SHN	AAM	GPS	20ppm	+/-0.04m
110	770947.425	8210969.562	0.688	Nail (non permanent)	TIP11	SHN	AAM	GPS	20ppm	+/-0.04m
111	770898.620	8210950.341	0.838	Nail (non permanent)		SHN	AAM	GPS	20ppm	+/-0.04m
121	757106.615	8222108.353	3.369	Nail (non permanent)	TIP12	SHN	AAM	GPS	20ppm	+/-0.04m
122 (Norsup)	757105.323	8222173.197	1.067	Pin in concrete (permanent)		SP / CONC	VLS	GPS	20ppm	+/-0.04m

Table 5 - Malekula Survey Control



SURVEY CONTROL - VANUATU LIDAR

UTM

Zone 58

Job No:	20076A	Job Name:	VANUATU LIDAR			Prepared by:	ST				
Date:	7/12/12	Location:	SANTO			Revision:	0				
AAM: AAM Group Pty Ltd VLS Vanuatu Land Survey PEL Pelydryn Pty Ltd VG Vanuatu Government			GPS: Static GPS TS: Total Station SL: Sprirt Leveling TG: Tide Gauge			BP: Brass Plaque SHN: Spring head Nail SP: Steel Pin GP: Ground Plate CONC: Concrete					
Station	ITRF2008 Coordinates		Height	Description	Location	Ground Mark	Installed By	Method (H/V)	Estimated		
	Easting	Northing	TG						H	V	
SANTO PRIMARY BASE (airport) +/-0.04m	738029.598	8284336.503	42.791	Corner of concrete wall (permanent)	airport	GP	VG	GPS	20ppm		
HOGB +/-0.04m	725964.931	8326186.519	3.029	pin in reef at shore (permanent)	north coast	SP / CONC	VLS	GPS	20ppm		
WHARF4 (saw1) 20ppm +/-0.04m	734894.896	8283453.280	1.750	pin in bitumen (permanent)	wharf	SP	PEL	GPS	20ppm		
WHARF5 (saw2) GPS +/-0.04m	734914.353	8283466.199	1.686	pin in bitumen (permanent)	wharf	SP	PEL	GPS	20ppm		
130 +/-0.04m	703400.827	8273895.254	26.510	Nail (non permanent)	TIP13	SHN	AAM	GPS	20ppm		
131 +/-0.04m	703420.381	8273986.881	27.166	Nail (non permanent)		SHN	AAM	GPS	20ppm		
140 +/-0.04m	713396.993	8275865.510	2.542	Nail (non permanent)	TIP14	SHN	AAM	GPS	20ppm		
141 +/-0.04m	713354.460	8275890.599	2.726	Nail (non permanent)		SHN	AAM	GPS	20ppm		
150 +/-0.04m	727018.008	8276432.947	11.695	Nail (non permanent)	TIP15	SHN	AAM	GPS	20ppm		
151 +/-0.04m	727047.744	8276390.453	10.336	Nail (non permanent)		SHN	AAM	GPS	20ppm		

TIP14

Table 6 - Santo Survey Control

For the full listing of control coordinates refer to '20076A Survey Control & TIP data 071212.xls'

7. TEST POINT SURVEY

The survey of test points included the capture of ground points at locations of various vegetation ground cover. This was to test the capability of LiDAR to penetrate vegetation and to resolve the correct ground surface.

In each list for TIP site dataset (refer '20076A Survey Control & TIP Data 071212.xls') ground shots have been measured on or under a range of vegetation types.



Figure 21 – Grassland



Figure 22 – Bushes and Shrub



Figure 23 – Woodland and Open Forest

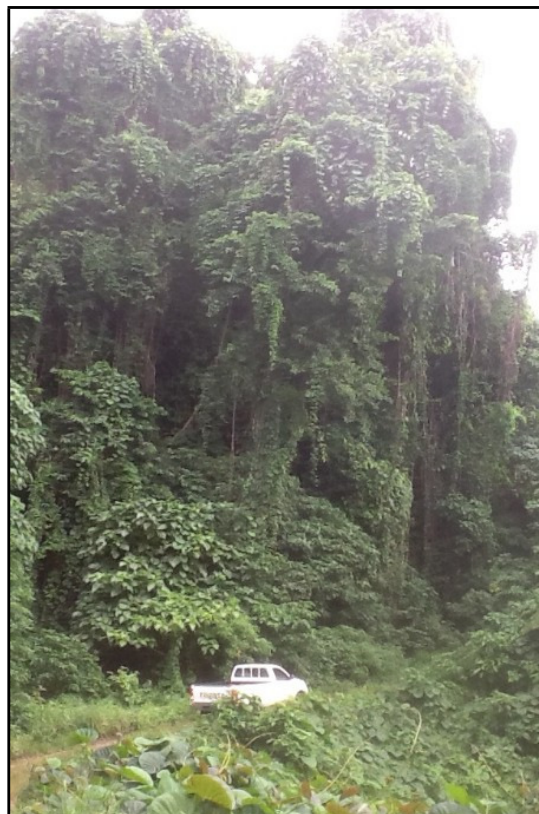


Figure 24 – Dense Vegetation

8. ACCURACIES

The accuracy of the TIP test point dataset is affected by a number factors including; survey methods, operator experience, and environment,

The reported accuracies are by estimation. On this project there have been a number of steps:

- GPS Control AAM
- Test Points AAM
- Tide Gauge Survey Pelydryn

Various stages and groups have been involved in the determinations of ground test point data which generally means there is an accumulation of errors.

8.1 GPS Control

A list of factors affecting the measurement and solution of control values include:

- Instrument Calibration
- Instrument Operating Specification
- Operators Experience
- Occupation Time
- Satellite Constellation

For this report the accuracy supplied here is only for the measured GPS data excluding the MSL values.

All control marks have the estimated accuracies:

- Horizontal 20ppm
- Vertical +/- 0.04m

8.2 Test Points (TIP)

The ground survey using conventional total station point observations includes:

- Instrument Calibration
- Instrument Operating Specification
- Operators Experience (Instrument and Prism)
- Accuracy of Survey Control
- Terrain (Grade and how a survey pole sits on the surface)

Adding the estimated GPS control accuracies, the estimated accuracy of the test points is:

- Horizontal 30ppm
- Vertical +/-0.04m – 0.06m

8.3 Accuracy Table

<u>Measurement Phase</u>	<u>Estimated Accuracy</u>	
	<u>Horizontal</u>	<u>Vertical</u>
Tide Gauge	N/A	TBA
Survey Control	20ppm	+/-0.04m
Test Points	30ppm	+/-0.04m – 0.06m

Table 7 - Accuracy Table